Experiments for Smooth RS-MCA Proximity Thresholds

Fresh-Eyes Threshold Memo, Missing Theorems, and Retained Locator Ledger

RS-MCA experimental notes

June 26, 2026

Abstract

This replacement starts with the current "fresh eyes" synthesis for the smooth Reed–Solomon Proximity Prize problem. The negative side is now quite rigid: Paper D v5 gives a prize-facing upper cap on the MCA threshold through a self-contained deep-point route, while Paper A and the list lower bounds rule out any no-slack capacity theorem. The positive side is not yet a theorem; it should be organized around two local-limit problems, L1 arbitrary-word locator fibers and M1 all-line residue packing, plus extension-line MCA, Paper D conversion/radius audits, and exact protocol-ledger rewrites.

The main new additions over the old `experiments.tex` are two compact bridge lemmas. First, for ordinary column-distance interleaving, interleaved lists inject into common feasible supports, so a uniform base locator-fiber theorem gives the same polynomial exponent for all protocol arities; there is no automatic $n^{\mu B}$ product loss. Second, a list bound plus an MCA bound implies a concrete line-decoding bound with loss $M + nL + 1$. The older BCHKS provenance notes, quotient-locator dictionaries, normal forms, arithmetic gates, and finite-threshold infrastructure are retained below whenever they still look useful.

Contents

1 Fresh-Eyes Settlement Memo

I would frame the current state as follows. The attached suite substantially settles the negative side and gives a plausible corrected theorem shape, but it does not yet settle the Proximity Prize threshold in the sharp "largest δ^*" sense. The clean prize-relevant statement is no longer "prove smooth RS reaches capacity". It is

$$\delta = 1 - \rho - \eta, \qquad \eta \text{ must clear generated-field entropy, quotient-core, MCA-floor,}$$

$$\text{interleaving, line-field, and challenge-field budgets.}$$

For

$$C = \mathrm{RS}[\mathbb{F}, D, k], \qquad n = |D|, \qquad \rho = k/n \in \{1/2, 1/4, 1/8, 1/16\},$$

the official challenge statements use this MCA/list-threshold form; in this experimental ledger the concrete prize envelope is [1]

$$\varepsilon_* = 2^{-128}, \qquad k \le 2^{40}, \qquad |\mathbb{F}| < 2^{256}.$$

The protocol-relevant soundness contribution has the schematic form

$$\varepsilon_{\mathrm{mca}}(C, \delta) + \frac{|\Lambda(C^{\equiv \mu}, \delta)|}{q_{\mathrm{line}}} + \text{query error.}$$

Thus a list theorem alone is not enough unless it is connected to the exact MCA, correlated-agreement, line-decoding, or curve-MCA quantity used by the protocol, and unless the denominator is the actual line or challenge field.

1.1 Bottom line

For the **MCA/correlated-agreement threshold**, the strongest attached result is Paper D v5's universal cap. Through the self-contained deep-point route and the stated dyadic-divisor hypotheses,

$$\delta^*_{\mathrm{MCA}}(2^{-128}) \le 1-\rho-2^{-9} \quad (\rho = 1/2, 1/4, 1/8),$$

and

$$\delta^*_{\mathrm{MCA}}(2^{-128}) \le 1-\rho-2^{-10} \quad (\rho = 1/16).$$

Equivalently, throughout the relevant unsafe intervals Paper D gives

$$\varepsilon_{\mathrm{mca}}(C,\delta) > \frac{1}{2k}\left(1-\frac{n}{|\mathbb{F}|}\right) \ge 2^{-86} \gg 2^{-128}.$$

This is a real prize-facing cap, not merely a heuristic obstruction. It is not an error-one theorem; it proves failure probability on the order of $1/k$, which is already far above 2^{-128}, but it does not characterize the full failure curve.

For the **interleaved-list threshold**, Paper A's base list lower bound plus the base-to-interleaved monotonicity gives a clean negative cap. At quotient scale $N \mid n$, with $\rho N \in \mathbb{Z}$, there is a received word with at least

$$\binom{N}{\rho N+1}/q, \qquad q = |\mathbb{F}|,$$

base-code codewords at radius $1-\rho-1/N$. Therefore the same radius is unsafe for the interleaved-list budget whenever this exceeds $2^{-128}q$. In the worst case $q < 2^{256}$, the following choices suffice:

ρ	N	list-failure radius	$\log_2\binom{N}{\rho N+1} - 256$	margin over 2^{128}
1/2	512	$1/2 - 2^{-9}$	251.17	123.17
1/4	512	$3/4 - 2^{-9}$	156.33	28.33
1/8	1024	$7/8 - 2^{-10}$	298.68	170.68
1/16	2048	$15/16 - 2^{-11}$	433.89	305.89

The fourth column is the exponent of the lower-bound list size after dividing by the worst-case field size 2^{256}; the fifth subtracts the target exponent 128.

1.2 What is already proved or very close to proved

The old no-slack smooth-domain capacity claim is dead. The quotient-locator construction gives explicit bad slopes for canonical lines and, over deployed smooth prime fields such as BabyBear, KoalaBear, and $3\cdot 2^{30}+1$, yields support-wise line-MCA error one on intervals starting at $1-\rho-2^{-18}$ for all four prize rates, with the sharper 2^{-17} gap at rates 1/2 and 1/4. These statements rule out any theorem saying that every radius below $1-\rho$ has negligible MCA error.

Paper D is the most prize-relevant negative theorem because it converts a deep point/list phenomenon into MCA at the exact 2^{-128}-style threshold. In v5, the headline MCA cap is routed locally through a self-contained deep-point argument rather than the Crites–Stewart list-to-agreement import. The remaining CA/list comparison and fallback statements still need their own radius, normalization, augmented-code, and import audits.

The list lower-bound side is also stronger than it is currently advertised. It should be promoted to an explicit interleaved-list negative corollary: base lists embed into ordinary interleaved lists, and the binomial numerators above already exceed the $2^{-128}|\mathbb{F}|$ budget in the full field envelope for the displayed quotient divisors.

1.3 The corrected positive theorem shape

The positive theorem should use

$$a = \lceil (1-\delta)n \rceil = k + \sigma, \qquad \delta = 1 - \rho - \eta,$$

and require at least

$$\sigma \log_2 q_{\text{gen}} \geq (1+\varepsilon) \log_2 \binom{n}{k+\sigma}, \qquad \sigma \geq C \frac{n}{\log n},$$

plus explicit quotient-profile and failure-floor ledgers. The first inequality uses the generated field q_{gen}, not a larger extension or challenge field that merely appears later in the protocol. The MCA numerator must be divided by the actual line field q_{line}.

The intended list target is an arbitrary-word locator-fiber theorem

$$\sup_U |\operatorname{Fib}_U(k+\sigma)| \leq n^B.$$

The intended MCA target is an all-line bound of the form

$$\varepsilon_{\text{mca}}(C, 1-\rho-\eta) \leq \frac{n^{1+o(1)} + 2^{(\beta(\rho)/H_2(\rho))\mathcal{Q}_H(\eta)(1+o(1))}}{q_{\text{line}}},$$

where quotient-periodic classes are removed or charged by $\mathcal{Q}_H(\eta)$. The current packet contains important special cases and reductions, but not these full arbitrary-word and all-line theorems.

1.4 What remains missing

1. **L1: arbitrary-word locator local limit.** This is the real list upper-bound theorem. Characteristic-zero rigidity, monomial-prefix lanes, and split-prime transfer results are useful but do not yet imply the uniform polynomial-field theorem needed for arbitrary received words.

2. **M1: all-line aperiodic residue packing.** Canonical-line exactness and quotient floors are lower-bound tools, not all-line upper bounds. The positive MCA theorem needs a fixed-field, all-line, all-base residue-packing bound.

3. **F1: extension-line MCA.** The list side has a clean extension-code identity. MCA does not. Base-valued bad lines can deflate when sampled over an extension field, while Paper D's cap indicates that genuinely extension-valued bad lines exist in relevant regimes. Any extension-line lift must therefore be a corrected-reserve theorem, not a blanket subfield statement.

4. **Paper D conversion audit.** Paper D v5's headline MCA cap is no longer load-bearing on the direct Crites–Stewart conversion. The self-contained deep-point step, support-wise monotonicity, integer-radius conventions, and the remaining CA/list import statements should still be independently checked.

5. **Protocol-ledger rewrite.** Deployed FRI/WHIR/IOP analyses must be rewritten with exact terms

   $$|\Lambda(C^{\equiv\mu}, \delta)|/q_{\text{line}} \quad \text{and} \quad \varepsilon_{\text{mca}} \text{ or line-decoding over the actual line field.}$$

   The survey toy protocol already has this shape; deployed analyses still need the same ledger.

2 New Bridge I: Common-Support Interleaved Fiber Injection

Status: New-local wrapper. The older ledger treated interleaving conservatively through a safe product bound $L_\mu(\delta) \le L_1(\delta)^\mu$. For ordinary column-distance RS interleaving, the following support-level injection gives the sharper upper bound: once the one-row locator-fiber theorem is proved, interleaving costs no extra exponent.

For $S \subseteq D$, let

$$L_S(X) = \prod_{x \in S} (X - x).$$

For a received word $U \in \mathbb{F}^D$, the condition $\deg(U \bmod L_S) < k$ means that the unique polynomial of degree less than $|S|$ interpolating $U|_S$ has degree less than k.

Definition 1 (one-row and common interleaved fibers)**.** *Let $C = \mathrm{RS}[\mathbb{F}, D, k]$, let $a \ge k$, and let $\mathbf{U} = (U_1, \dots, U_\mu) \in (\mathbb{F}^\mu)^D$. Define*

$$\mathrm{Fib}^{(\mu)}_{\mathbf{U}}(a) = \{S \subseteq D : |S| = a,\ \deg(U_i \bmod L_S) < k \text{ for every } i = 1, \dots, \mu\}.$$

For one row, write $\mathrm{Fib}_{U_i}(a)$ for the same condition with only U_i.

Lemma 1 (common-support interleaved fiber injection)**.** *Let $C = \mathrm{RS}[\mathbb{F}, D, k]$, $n = |D|$, and let $a = \lceil (1-\delta)n \rceil \ge k$. For every received μ-row word $\mathbf{U} \in (\mathbb{F}^\mu)^D$, ordinary column-distance interleaving satisfies*

$$|\Lambda(C^{\equiv\mu}, \delta, \mathbf{U})| \le |\operatorname{Fib}^{(\mu)}_{\mathbf{U}}(a)| \le \min_{1 \le i \le \mu} |\operatorname{Fib}_{U_i}(a)|.$$

Consequently,

$$\sup_{\mathbf{U}} |\Lambda(C^{\equiv\mu}, \delta, \mathbf{U})| \le \sup_{U} |\operatorname{Fib}_U(a)|.$$

In particular, if L1 proves $\sup_U |\operatorname{Fib}_U(k+\sigma)| \le n^B$, then

$$|\Lambda(C^{\equiv\mu}, 1 - \rho - \sigma/n)| \le n^B$$

for every ordinary column-distance protocol arity μ.

Proof. Fix a total order on D. A listed interleaved codeword $(P_1, \dots, P_\mu) \in C^{\equiv\mu}$ at column distance at most δ from $\mathbf{U}$ agrees with $\mathbf{U}$ on at least a common columns. Choose the first a-subset S of those agreeing columns. For this S, every row $U_i|_S$ is interpolated by the degree-$< k$ polynomial P_i, hence $S \in \operatorname{Fib}^{(\mu)}_{\mathbf{U}}(a)$.

For fixed S, the tuple $(P_1, \dots, P_\mu)$ is uniquely determined, because each P_i has degree less than k and is fixed by its values on $|S| \ge k$ points. Thus the canonical-support map from listed interleaved codewords to common feasible supports is injective. The second inequality is immediate because every common feasible support is feasible for each individual row. Taking suprema gives the final claim. □

Remark 1 (impact)**.** *For ordinary column-distance RS interleaving, L2 should be demoted from a separate product-exponent problem to a corollary of L1. The hard list theorem remains the base arbitrary-word locator local limit.*

3 New Bridge II: List plus MCA Gives Line Decoding

Status: New-local bridge. The next proposition turns a line-decoding proof obligation into a finite-constant budget once L1 and M1 are available. It uses finite slopes and the support-wise MCA normalization used in this packet.

Proposition 1 (list/MCA bridge to line decoding)**.** *Let $C = \mathrm{RS}[\mathbb{F}, D, k]$, let $|D| = n$, and let $a = \lceil (1-\delta)n \rceil > k$. Suppose that every allowed received line $f + \gamma g$ has at most M support-wise MCA-bad finite slopes at radius δ, equivalently $\varepsilon_{\mathrm{mca}}(C, \delta) \le M/|\mathbb{F}|$ for the chosen finite slope sampler. Suppose also that the two-row interleaved list size satisfies*

$$|\Lambda(C^{\equiv 2}, \delta, W)| \le L \qquad \text{for every } W \in (\mathbb{F}^2)^D.$$

Then C is line-decodable with parameters

$$(\delta,\ M + nL + 1,\ n + 1)$$

in the following sense: if an allowed received line $f + \gamma g$ has more than $M + nL$ finite slopes γ that are δ-close to selected codewords $U(\gamma) \in C$, then there are codewords $c_1, c_2 \in C$ such that

$$U(\gamma) = c_1 + \gamma c_2$$

for at least $n + 1$ of those slopes.

Proof. For each close slope γ, choose a support $S_\gamma \subseteq D$ of size a on which $f + \gamma g$ agrees with the selected codeword $U(\gamma)$. Discard the at most M support-wise MCA-bad slopes. For every remaining slope, support-wise containment gives codewords $c_{1,\gamma}, c_{2,\gamma} \in C$ such that

$$c_{1,\gamma}|_{S_\gamma} = f|_{S_\gamma}, \qquad c_{2,\gamma}|_{S_\gamma} = g|_{S_\gamma}.$$

Thus $(c_{1,\gamma}, c_{2,\gamma})$ is a two-row interleaved codeword δ-close to the received two-row word (f, g), using the common support S_γ. There are at most L possible pairs.

If the original line has more than $M + nL$ close slopes, then after discarding bad slopes more than nL slopes remain. By pigeonhole, one pair (c_1, c_2) occurs for at least $n + 1$ slopes. For each such slope, the codewords $U(\gamma)$ and $c_1 + \gamma c_2$ both agree with $f + \gamma g$ on S_γ, hence agree with each other on $a > k$ positions. Since both have degree less than k, they are identical codewords. This proves the stated line-decoding conclusion. □

Remark 2 (finite-constant consequence)**.** *If M1 gives $M = n^{1+o(1)}$ and L1 plus Lemma 1 gives $L = n^B$, then Proposition 1 gives*

$$M + nL + 1 \le n^{B+1+o(1)}.$$

For $n \le 2^{44}$ and $|\mathbb{F}| < 2^{256}$, a raw list numerator n^B is compatible with a 2^{-128} field budget only for about $B \lesssim 2.9$, and the line-decoding bridge is compatible only for about $B \lesssim 1.9$. Thus the bridge is useful, but it leaves a real finite-constant target.

4 Updated Positive-Theorem Package

The clean theorem to aim for is the following finite-reserve package. Let

$$C = \mathrm{RS}[\mathbb{F}_{\mathrm{line}}, D, k], \qquad n = |D|, \qquad a = k + \sigma, \qquad \delta = 1 - a/n = 1 - \rho - \eta.$$

Assume

$$\sigma \log_2 q_{\mathrm{gen}} \geq (1+\varepsilon) \log_2 \binom{n}{k+\sigma}, \qquad \sigma \geq C_{\rho,B,\varepsilon} \frac{n}{\log n},$$

the quotient profile $\mathcal{Q}_H(\eta)$ is empty or explicitly budgeted, η lies above all proved failure floors including Paper D's universal-cap floor, and the MCA or line experiment is proved over q_{line}, not merely over q_{chal}.

Conjecture 1 (corrected finite reserve package)**.** *Under the preceding hypotheses, the desired conclusions are*

$$|\Lambda(C^{\equiv \mu}, \delta)| \leq n^B$$

for all ordinary column-distance protocol arities μ, by Lemma 1; and

$$\varepsilon_{\mathrm{mca}}(C, \delta) \leq \frac{n^{1+o(1)} + 2^{(\beta(\rho)/H_2(\rho))\mathcal{Q}_H(\eta)(1+o(1))}}{q_{\mathrm{line}}},$$

or the same bound over q_{gen} plus a proved extension-line lift. If the consumed primitive is line decoding, then Proposition 1 should give

$$C \textit{ is } (\delta,\ M + nL + 1,\ n+1) \textit{ line-decodable},$$

with M and L supplied by M1 and L1.

This package would turn the current negative cap into a sharp threshold program: below the cap one cannot hope for 2^{-128} MCA soundness, while above the corrected reserve one would have finite numerators that can be checked against q_{line}.

5 Immediate Experimental Additions

1. Add `experimental/interleaved_fiber_injection.md`, containing Lemma 1. Update certificate logic so an L1 locator-fiber theorem gives $L_\mu = n^B$, not $n^{\mu B}$, for ordinary column-distance RS interleaving.
2. Add `experimental/list_mca_to_linedecoding.md`, containing Proposition 1. This gives a concrete M2 bridge and turns line decoding into a finite-constant ledger problem.
3. Build a certificate scanner reporting generated-field entropy margin, quotient profile after $k = \rho n - r$, Paper D cap floor, base and interleaved list numerators, MCA numerator, line-decoding numerator, field separation among q_{gen}, q_{line}, and q_{chal}, extension status, and exact failure audit.
4. Keep CA/list statements that use Crites–Stewart marked CONDITIONAL until audited. Paper D v5's MCA cap should instead be tracked as PROVED when the divisor, binomial, and subfield hypotheses are verified.

6 Fresh Non-Claims

This revision does not claim an exact settlement of the smooth-case Proximity Prize threshold. It claims that the shape is now rigid:

$$\boxed{\text{No unslacked capacity theorem; prove a reserve theorem at } 1-\rho-\eta.}$$

The theorem-backed negative side gives a draft universal MCA cap at constant gap 2^{-9} for $\rho = 1/2, 1/4, 1/8$ and 2^{-10} for $\rho = 1/16$. The positive side should be pursued as

$$\text{L1 locator local limit} + \text{M1 residue-line packing} + \text{extension-line theorem} + \text{protocol-ledger rewrite.}$$

The interleaved-list problem is less fundamental than it looked in the old file: for column-distance interleaving, a uniform locator-fiber theorem immediately bounds the interleaved list at the same exponent.

7 Retained Previous Experiments Material

The remaining sections are retained from the previous `experiments.tex` whenever they still provide useful provenance, notation, normal forms, finite certificate gates, or non-claims. The new front matter above should be read as the current threshold-level guide; the older material below remains a toolbox.

8 Status, Provenance, and What Counts as New

This file is the place where we collect new things, but it must separate new repository contributions from imported constructions. The status labels used below are:

Imported	an external theorem, proof pattern, or conversion;
Wrapper	a repository repackaging, dictionary, or pigeonhole consequence;
New-local	a restricted algebraic theorem proved in this note;
Target	a precise proof obligation not yet promoted;
Heuristic	evidence or a search direction, not a theorem.

Hard attribution rule. Any locator-polynomial argument derived from the subgroup construction of Ben-Sasson–Carmon–Haböck–Kopparty–Saraf must cite [2, Thm. 7.1] where the construction is stated or proved. Any admissible subgroup instantiation must also cite [2, Thm. 1.13]. The citation belongs at the theorem statement and again at the proof entry point; a bibliography entry or acknowledgement alone is not enough.

What is new in this note. The following items are the repository-side contributions collected here.

- A smooth-quotient notation dictionary translating BCHKS subgroup notation into the RS-MCA notation $Q = D^{c_{\mathrm{fib}}}$ and the dyadic/FFT-domain ledgers used in Papers A–D.
- A list-fiber pigeonhole wrapper: once a locator construction assigns many subset certificates to a smaller slope set, one heavy slope has many locator certificates, and, after a distinctness check, many nearby codewords.

- A slack-two and subfield-pigeonhole proof target for the older Crites–Stewart/CA route: the imported locator construction should be run at the augmented-code rung needed by that conversion, and slopes should be pigeonholed over the generated/base field when the construction really takes values there.

- A New-local divisibility gate in the restricted Paper B window $B = \mathbb{F}_p$, $F = \mathbb{F}_{p^2}$, $D = \mathbb{F}_p$, $t = \sigma = 2$, $j = 3$. This is independent of the BCHKS priority issue and is the main theorem proved in Section 14.

What is not new. The polynomial identity behind the item-(d) locator proof is not new here. In BCHKS notation it is the expansion of a vanishing polynomial over a complement of a chosen subset; in smooth-quotient notation it is the same expansion of $\prod_{b\in A}(X^{c_{\text{fib}}} - b)$, up to choosing a subset or its complement. The use of ℓ for the chosen subset size also follows the external presentation closely enough that it should either be cited or renamed to ℓ_{pg}.

Source map. The main papers remain the authoritative documents.

- Paper A, `tex/RS_disproof_v3.tex`, refutes the old no-slack smooth-domain support-wise line-MCA statement. If it uses the locator construction, it should cite BCHKS and call the repository contribution a smooth-domain/list-fiber corollary, not a new locator proof.

- Paper B, `tex/slackMCA_v3.tex`, is the main target for the local resonance material in Sections 11–17.

- Paper C, `tex/snarks_v4.tex`, records protocol ledgers. Nothing in this note may be consumed by a protocol without a separate conversion theorem.

- Paper D, `tex/cs25_cap_v4.tex`, gives the universal cap via an imported list-to-agreement route. Its locator-side input should cite BCHKS explicitly if it uses the quotient-locator construction.

9 BCHKS Locator Construction in Smooth-Quotient Notation

Status: Imported plus Wrapper. This section is a provenance-safe restatement and dictionary. It is included so later main-paper edits have a clean place to copy from.

9.1 Imported theorem/proof pattern

BCHKS prove a general multiplicative-subgroup construction [2, Thm. 7.1]. In a compact form, the imported statement is as follows.

Theorem 1 (BCHKS quotient-locator construction, imported)**.** *Let $G \subseteq H \subseteq \mathbb{F}_q^*$ be multiplicative subgroups and let $\Phi : H \to G$ be $x \mapsto x^c$, where $c = |H|/|G|$. Let $E \subseteq G$, and let $\ell_{\text{pg}}, a \geq 1$ satisfy*

$$|E^{(+\ell_{\text{pg}})}| = \#\{e_1 + \cdots + e_{\ell_{\text{pg}}} : e_i \in E \text{ distinct}\} \geq a.$$

Let $D = \Phi^{-1}(E)$, $n = |D|$, and

$$C = \mathrm{RS}[\mathbb{F}_q, D, n - (\ell_{\text{pg}} + 2)c].$$

Then, for $\gamma = \ell_{\mathrm{pg}} c/n$, there are functions $f, g : D \to \mathbb{F}_q$ such that

$$\#\{z \in \mathbb{F}_q : \Delta(f + zg, C) \leq \gamma\} \geq a,$$

while

$$\Delta([f, g], C^2) \geq (\ell_{\mathrm{pg}} + 1)c/n.$$

Reference. This is exactly the imported construction of [2, Thm. 7.1]; the proof is not reproduced as a repository contribution. BCHKS define the line endpoints by the monomials $x^{n-\ell_{\mathrm{pg}}c}$ and $x^{n-(\ell_{\mathrm{pg}}+1)c}$, then expand a vanishing polynomial over the complement of a chosen ℓ_{pg}-subset. The coefficient of $X^{n-(\ell_{\mathrm{pg}}+1)c}$ is an affine transform of a restricted sum from $E^{(+\ell_{\mathrm{pg}})}$, giving the exceptional slope values. □

The BCHKS Theorem 1.13 is the admissible-subgroup specialization of this construction: take $E = G$, take $H = D$, set $c = |D|/|G|$, and use $\ell_{\mathrm{pg}} = |G|/2$. Any RS-MCA statement corresponding to that specialization should cite [2, Thm. 1.13] directly.

9.2 Notation dictionary

BCHKS notation	**RS-MCA notation**	**Meaning**
H	evaluation domain D	Large multiplicative domain on which codewords are evaluated.
G	quotient $Q = D^{c_{\mathrm{fib}}}$	Image of the power map; one point in Q has c_{fib} preimages in D.
$c = \|H\|/\|G\|$	$c_{\mathrm{fib}} = n/\|Q\|$	Fiber size of the quotient map. Avoid calling this a, because BCHKS also uses a for the number of exceptional slopes.
$\Phi(x) = x^c$	$x \mapsto x^{c_{\mathrm{fib}}}$	Quotient/fiber map.
$E \subseteq G$	active quotient subset $E \subseteq Q$	Quotient-level support used by the locator.
$E^{(+\ell_{\mathrm{pg}})}$	restricted first-coefficient/slope set	Sums of distinct chosen quotient elements; slopes are affine transforms of these sums.
ℓ_{pg}	ℓ_{pg} or renamed subset size	Number of chosen quotient elements. Use ℓ_{pg} in RS-MCA text to avoid collision with other ℓ's.
$f = x^{n-\ell_{\mathrm{pg}}c}$, $g = x^{n-(\ell_{\mathrm{pg}}+1)c}$	locator line endpoints	Same monomial line after quotient notation is substituted.
$P_S(X) = \prod_{\alpha \in S'}(X^c - \alpha)$	$\prod_{b \in A}(X^{c_{\mathrm{fib}}} - b)$, possibly after complementing A	The locator polynomial.
exceptional z's	bad slopes / heavy list fibers	Slope parameters for which a line word is close to the RS code.

9.3 The shared locator identity

The following identity is written down because it is the exact place where priority can be lost. The identity is the BCHKS identity in RS-MCA notation.

Lemma 2 (Locator expansion, imported identity)**.** *Let D map c_{fib}-to-one onto a quotient set E by $x \mapsto x^{c_{\mathrm{fib}}}$, and let $A \subseteq E$ have size ℓ_{pg}. Put $A^c = E \setminus A$ and define*

$$P_A(X) = \prod_{\alpha \in A^c} (X^{c_{\mathrm{fib}}} - \alpha).$$

Then

$$P_A(X) = X^{n-\ell_{\mathrm{pg}} c_{\mathrm{fib}}} - e_1(A^c) X^{n-(\ell_{\mathrm{pg}}+1)c_{\mathrm{fib}}} + \textit{terms of degree at most } n - (\ell_{\mathrm{pg}} + 2)c_{\mathrm{fib}}.$$

Equivalently, since $e_1(A^c) = e_1(E) - e_1(A)$, the coefficient of the next monomial is an affine transform of the restricted sum $e_1(A)$.

Reference. This is the elementary expansion used inside [2, Thm. 7.1]. We record it only to fix notation. It should not be cited as a new lemma of the repository unless the BCHKS provenance is present at the statement. □

9.4 List-fiber pigeonhole wrapper

The next proposition is the repository-side wrapper around the imported identity. It is intentionally stated in certificate form, because distinctness of the nearby codewords is a separate check in concrete instantiations.

Proposition 2 (List-fiber pigeonhole wrapper)**.** *Let $\mathcal{A}$ be a family of ℓ_{pg}-subsets of a quotient set E. Suppose the BCHKS locator construction assigns to each $A \in \mathcal{A}$ a slope $z(A)$ lying in a finite set S, and a locator certificate showing that the corresponding word $f + z(A)g$ is within radius γ of a locator codeword $P_A \in C$. Then some slope $z_0 \in S$ has at least*

$$\frac{|\mathcal{A}|}{|S|}$$

locator certificates above it. If the codewords P_A attached to the certificates over z_0 are distinct, then the Hamming ball of radius γ around $f + z_0 g$ contains at least $|\mathcal{A}|/|S|$ codewords of C.

Proof. The first assertion is the pigeonhole principle applied to the map $A \mapsto z(A)$. The second assertion is only the definition of list size, after the distinctness check removes possible duplicate locator codewords. The construction that supplies the certificates is imported from Theorem 1. □

Remark 3 (What the wrapper adds)**.** *The new part here is not the locator polynomial. The wrapper isolates the extra bookkeeping needed by RS-MCA: which slope set pays the pigeonhole bill, whether the slope set is the ambient field $\mathbb{F}$ or a generated/base subfield B, and whether locator certificates remain distinct after specialization. Those are the checks needed before using the construction in Papers A, B, or D.*

9.5 Slack-two and subfield target for the older CS25 route

Status: Target. The older CS25/CA route needs a locator-heavy list fiber in the augmented code rung $C^+ = \mathrm{RS}[\mathbb{F}, D, k+1]$, because that is the object fed into the Crites–Stewart list-to-agreement conversion. Paper D v5 no longer uses this as the load-bearing MCA-cap route, but the target remains useful for CA/list comparison audits. The proof obligation should be stated as follows.

Proposition 3 (Slack-two/subfield locator target)**.** *Let $D \subseteq B \subseteq \mathbb{F}$, and suppose a BCHKS-type quotient locator construction over D produces slopes in B, not merely in $\mathbb{F}$. If the one-rung/augmented-code version produces distinct locator codewords in $C^+ = \mathrm{RS}[\mathbb{F}, D, k+1]$ for a family $\mathcal{A}$ of certificates, then there exists a slope $z_0 \in B$ with list size at least*

$$\frac{|\mathcal{A}|}{|B|}$$

inside the appropriate Hamming ball for C^+.

Bookkeeping only. This is Proposition 2 with $S = B$. The nontrivial work is external or elsewhere: one must verify the BCHKS locator hypotheses at the augmented-code rung, verify that the slope coefficients lie in B, and verify distinctness of the produced codewords. Until those checks are present, this proposition is a target, not a promoted Paper D theorem. □

9.6 Pasteable main-paper wording

Paper A wording. The polynomial locator identity used here is the quotient-fiber form of the construction of Ben-Sasson–Carmon–Haböck–Kopparty–Saraf [2, Thm. 7.1]; their Theorem 1.13 is the admissible subgroup instantiation. Thus the locator proof itself is not a new contribution of this manuscript. The present use is the smooth-domain specialization, the field/divisor bookkeeping, and the list-fiber pigeonhole packaging.

Paper D wording. The locator-side input is the BCHKS quotient-locator construction [2, Thm. 7.1; cf. Thm. 1.13], written in the smooth-domain quotient notation $Q = D^{c_{\mathrm{fib}}}$. The additional bookkeeping needed here is the slack-two augmented-code rung and, when available, subfield pigeonholing over B rather than over the ambient extension field $\mathbb{F}$.

10 Triage of the AI-Generated Four-Item Packet

Status: Audit. The four advertised items should be positioned as follows before any promotion into a main paper. The original AI packet used the labels (a)–(d) without making them self-contained in this note, so we decode them here.

Item	Short name	Meaning in this ledger
(a)	Weak-slack positive regime	A positive-looking proximity/list or MCA-adjacent statement that only reaches slack on the order of $1/\sqrt{n}$. It is useful as a comparison theorem, but it is far from the constant/reserve bookkeeping needed for the corrected RS-MCA target.
(b)	Finite Fermat-prime packet	A special finite-domain instantiation over only three Fermat-prime cases, in the current experimental ledger represented by the $p \in \{17, 257, 65537\}$ style checks. It is evidence or a worked computation, not a uniform theorem.
(c)	Exponential-field construction	A large-field variant whose proof requires the ambient field to be exponential in the domain/parameter size. This can clarify the mechanism but does not match the polynomial-field or challenge-field regimes needed by the Proximity Prize ledger.
(d)	BCHKS quotient-locator packet	The interesting locator/proximity-gap construction: choose a quotient of a multiplicative subgroup, use ℓ_{pg}-subsets and restricted sums to create many exceptional slopes or nearby locator certificates. This is the BCHKS construction of [2, Thm. 7.1], with the admissible subgroup instance in [2, Thm. 1.13]; the repository contribution is only the smooth-domain dictionary, list-fiber wrapper, and later slack-two/subfield bookkeeping.

Item (d) should not be confused with the New-local divisibility-gate theorem in Section 14. The former is an imported locator/proximity construction; the latter is an independent restricted Paper B resonance calculation.

Item	Limitation	Repository handling
(a)	Works only at slack on the order of $1/\sqrt{n}$.	Use as a weak-slack comparison point only. It does not clear the corrected-reserve MCA objective.
(b)	Applies only to three Fermat-prime cases.	Treat as finite-prime evidence or a worked computation, not as a uniform smooth-domain theorem.
(c)	Requires exponential field size.	Treat as a large-field sanity check. It is not directly usable for polynomial-field or prize-ledger parameters.
(d)	Most relevant, but the locator proof is essentially BCHKS.	Import, cite, and adapt from [2, Thm. 7.1 and Thm. 1.13]. The repository contribution is only the smooth-domain/list-fiber/slack-two packaging after exact hypotheses are checked.

Promotion checklist. Before any statement using item (d) moves to Papers A–D, check the following.

- Does the statement cite BCHKS at the theorem and proof entry point?
- Is the object a proximity-gap statement, a list statement, CA, MCA, line-decoding, or a protocol-facing theorem?
- Which field pays the slope/list pigeonhole bill: B, $\mathbb{F}$, or another generated field?
- Are the locator codewords distinct after specialization?
- Is the claimed slack in the same normalization as the consumed theorem?
- Are items (a), (b), and (c) kept out of asymptotic/protocol claims unless their limitations are explicitly part of the theorem statement?

11 Restricted Resonance Setup for Paper B

Status: New-local for the divisibility gate; otherwise local normal-form bookkeeping. The remaining sections preserve and reorganize the Cycle 14–18 algebraic material. This branch is independent of the BCHKS priority correction.

The common restricted setting is

$$B = \mathbb{F}_p, \qquad F = \mathbb{F}_{p^2}, \qquad D = \mathbb{F}_p, \qquad t = \sigma = 2, \qquad j = 3,$$

and we work off the tangent/global endpoint

$$R_0 = \{ [W]_E \wedge [B_{\mathrm{num}}]_E = 0 \}.$$

The field ledgers remain distinct:

$$q_{\mathrm{gen}} = p, \qquad q_{\mathrm{line}} = p^2.$$

The regime is sub-reserve, since $\eta = \sigma/n = 2/n$. Thus none of the following proves a corrected-reserve list, CA, MCA, line-decoding, SNARK, or protocol theorem.

The base/generated field is $B = \mathbb{F}_p$. The line field is the quadratic extension $F = \mathbb{F}_{p^2} = B(\alpha)$, with basis $\{1, \alpha\}$ over B and $\alpha^2 = \nu \in B$. The domain is $D = \mathbb{F}_p$, so $n = |D| = p$.

A degree-two residue-line datum consists of

$$E \in F[X], \qquad \deg E = 2, \qquad E(x) \neq 0 \text{ for all } x \in D,$$

and a numerator

$$B_{\mathrm{num}} \in F[X], \qquad \deg B_{\mathrm{num}} < 2.$$

We write

$$A = F[X]/E, \qquad [P]_E \in A, \qquad b = [B_{\mathrm{num}}]_E.$$

The anchor word on D is interpolated by a polynomial W. The operation $u \wedge v$ means the determinant of the coordinate vectors of $u, v \in A$ in a chosen F-basis of the two-dimensional F-space A. Off R_0, the pair $\{[W]_E, b\}$ is such a basis.

The co-support has size $j = 3$. If $T \subset D$ is the three-point co-support, define

$$\tau_1 = e_1(T), \qquad \tau_2 = e_2(T), \qquad \tau_3 = e_3(T).$$

All valid co-supports have $\tau_i \in B$. The quantity C_2 denotes the number of distinct slopes produced by the relevant landing branch after restricting to valid split triples $T \subset \mathbb{F}_p$. Bounding landing points therefore bounds C_2, but a sharp slope theorem may require more because many landing points can share a slope.

12 Cycle 14 Normal Form

The Cycle 14 audit reduces the $t = 2, j = 3$ landing equation to two affine forms in the elementary co-support parameters

$$\tau = (\tau_1, \tau_2, \tau_3).$$

Lemma 3 (Affine wedge normal form)**.** *Assume $\{[W]_E, b\}$ is an F-basis of A. Then the landing equation can be written in the form*

$$\iota(\tau) = A_0(\tau_1, \tau_2) - \tau_3[W]_E, \qquad \mu(\tau) = B_0(\tau_1, \tau_2) - \tau_3 b,$$

where

$$A_0 = p_1[W]_E + p_2 b, \qquad B_0 = q_1[W]_E + q_2 b,$$

and $p_i, q_i \in F$ are affine-linear functions of (τ_1, τ_2). If the basis is normalized so $[W]_E \wedge b = 1$, then

$$\Delta = \iota \wedge \mu = (p_1 - \tau_3)(q_2 - \tau_3) - p_2 q_1.$$

Proof. The Cycle 14 quotient calculation, which is a concrete instance of Paper B's residue-line normal form, gives the displayed affine forms. Expanding in the ordered basis $\{[W]_E, b\}$, the coordinate vector of ι is $(p_1 - \tau_3, p_2)$, while the coordinate vector of μ is $(q_1, q_2 - \tau_3)$. Their wedge determinant is therefore

$$(p_1 - \tau_3)(q_2 - \tau_3) - p_2 q_1.$$

□

13 Cycle 18 Component Split

Choose a basis $F = B + \alpha B$ with $\alpha^2 = \nu \in B$, and write

$$p_1 = p_{10} + \alpha p_{11}, \quad p_2 = p_{20} + \alpha p_{21}, \quad q_1 = q_{10} + \alpha q_{11}, \quad q_2 = q_{20} + \alpha q_{21}.$$

Split

$$\Delta = \Delta_0 + \alpha \Delta_1, \qquad \Delta_0, \Delta_1 \in B[\tau_1, \tau_2, \tau_3].$$

Theorem 2 (Monic quadratic and linear alpha component)**.** *In the setting above,*

$$\begin{aligned}
\Delta_0 &= \tau_3^2 - (p_{10} + q_{20})\tau_3 + p_{10}q_{20} + \nu p_{11}q_{21} - p_{20}q_{10} - \nu p_{21}q_{11}, \\
\Delta_1 &= \ -(p_{11} + q_{21})\tau_3 + p_{10}q_{21} + p_{11}q_{20} - p_{20}q_{11} - p_{21}q_{10}.
\end{aligned}$$

Consequently Δ_0 is monic quadratic in τ_3, while Δ_1 has degree at most one in τ_3.

Proof. Substitute the coordinate expressions for p_i, q_i into the identity of Lemma 3 and use $\alpha^2 = \nu$. Collecting the coefficient of 1 gives the displayed formula for Δ_0, and collecting the coefficient of α gives the displayed formula for Δ_1. The coefficient of τ_3^2 in Δ_0 is 1, and the α-component has no τ_3^2-term. □

Corollary 1 (Graph reduction outside the zero-alpha branch)**.** *If Δ_1 is not the zero polynomial, write*

$$\Delta_1 = s(\tau_1, \tau_2)\tau_3 + h(\tau_1, \tau_2).$$

On the open locus $s \neq 0$, the common-zero condition $\Delta_0 = \Delta_1 = 0$ is supported on the graph

$$\tau_3 = -h/s.$$

Proof. This is immediate from solving the linear equation $\Delta_1 = 0$ for τ_3 on $s \neq 0$. □

14 New Divisibility-Gate Theorem

The next theorem is the main New-local theorem extracted from the Cycle 18 reduction. It is stronger than merely saying that the branch is a graph: it shows that any two-dimensional family must satisfy an explicit divisibility identity.

Theorem 3 (Divisibility gate for the graph branch)**.** *Assume the restricted setting above and assume $\Delta_1 \neq 0$. Write*

$$\Delta_1 = s(\tau_1, \tau_2)\tau_3 + h(\tau_1, \tau_2),$$

and define the denominator-cleared graph polynomial

$$G(\tau_1, \tau_2) = s(\tau_1, \tau_2)^2 \Delta_0(\tau_1, \tau_2, -h(\tau_1, \tau_2)/s(\tau_1, \tau_2)).$$

Then $\deg G \leq 4$. If G is not the zero polynomial, then

$$\#\{\, \tau \in B^3 : \Delta_0(\tau) = \Delta_1(\tau) = 0 \,\} = O(p).$$

Consequently the corresponding number of distinct slopes satisfies

$$C_2 = O(p) = O(n).$$

Therefore a growing-prime family with

$$C_2 = \Theta(p^2) = \Theta(q_{\text{line}})$$

in the nonzero-Δ_1 branch can occur only if $G \equiv 0$.

Proof. Because the p_i, q_i are affine-linear in (τ_1, τ_2), the coefficient s has degree at most 1, and h has degree at most 2. Since Δ_0 is monic quadratic in τ_3, clearing the denominators after substituting $\tau_3 = -h/s$ gives a polynomial G of degree at most 4.

On $s \neq 0$, the equation $\Delta_1 = 0$ forces $\tau_3 = -h/s$. Hence every common zero on this open locus gives a base pair (τ_1, τ_2) with $G(\tau_1, \tau_2) = 0$. If G is nonzero, Schwartz–Zippel gives at most $4p$ such base pairs, and each base pair gives at most one value of τ_3.

It remains to count the exceptional locus $s = 0$. There $\Delta_1 = 0$ also forces $h = 0$. If s is a nonzero affine form, this is contained in an affine line; if $s \equiv 0$, then h is a nonzero polynomial of degree at most 2. In either case there are $O(p)$ base pairs. For each such pair, Δ_0 is a monic quadratic in τ_3, so it gives at most two values of τ_3. The total common-zero count is $O(p)$.

Finally, the number of distinct slopes is bounded by the number of landing points τ. Thus $C_2 = O(p)$ whenever $G \not\equiv 0$, and any $\Theta(p^2)$ family in this branch must satisfy $G \equiv 0$. □

Remark 4. *The identity $G \equiv 0$ says that, after clearing the s-denominator, Δ_0 is divisible by the linear equation Δ_1 on the graph branch. This is the first exact algebraic gate that a large slope family in this restricted window must pass.*

15 Nearby Proven Gates

The same restricted lane contains two other useful gates. They are included here because they explain where Theorem 3 fits.

Proposition 4 (Base-component complete-intersection gate)**.** *Suppose $\Delta_0, \Delta_1 \in B[\tau_1, \tau_2, \tau_3]$ have no common surface component. Then*

$$\#V(\Delta_0, \Delta_1)(B) = O(p),$$

and hence this branch contributes $C_2 = O(p)$ slopes.

Proof. The two components have bounded degree. If they share no surface component, their common zero set in $\mathbb{A}^3_B$ is a curve of bounded degree. A bounded-degree affine curve over $\mathbb{F}_p$ has $O(p)$ rational points, and slopes are bounded by landing points. ☐

Proposition 5 (Cycle 16 determinant gate)**.** *Let $Q(z_0, z_1)$ be the Cycle 15–16 determinant consistency polynomial for the affine system*

$$L_z(\tau) = \iota(\tau) - z\mu(\tau) = 0, \qquad z = z_0 + \alpha z_1.$$

If Q is not identically zero, then in the restricted $D = \mathbb{F}_p$, $t = \sigma = 2$, $j = 3$, off-R_0 window,

$$C_2 \leq 4p = O(p) = O(n).$$

Proof. Cycle 16 gives $\deg Q \leq 4$ and shows that every landing slope must satisfy $Q(z_0, z_1) = 0$. If $Q \neq 0$, Schwartz–Zippel over $B = \mathbb{F}_p$ gives at most $4p$ possible pairs (z_0, z_1), hence at most $4p$ slopes. ☐

16 Slope Map and Remaining Heuristics

On the graph branch, the landing equation $\iota = z\mu$ gives

$$p_1 - \tau_3 = zq_1, \qquad p_2 = z(q_2 - \tau_3).$$

Where $q_1 \neq 0$, substituting $\tau_3 = -h/s$ yields

$$z(\tau_1, \tau_2) = \frac{p_1 + h/s}{q_1}.$$

Equivalently, eliminating τ_3 recovers the Cycle 14 quadratic

$$q_1 z^2 - (p_1 - q_2)z - p_2 = 0,$$

so the residual map is controlled by

$$(\tau_1, \tau_2) \longmapsto [\, q_1 : (p_1 - q_2) : p_2 \,] \in \mathbb{P}^2(F).$$

Heuristic 1 (Exceptional identity should be rare)**.** *For generic source data, the polynomial G should not vanish identically. Thus the graph branch should usually be curve-sized, and the first possible large-slope families should come from algebraically special data satisfying the divisibility identity $G \equiv 0$ or from the zero-alpha branch $\Delta_1 \equiv 0$.*

Heuristic 2 (What remains after the gate)**.** *If $G \equiv 0$, the problem is no longer to show that the common-zero set is small. It may be surface-sized. The remaining question is whether the projective coefficient map has one-dimensional image on every source-valid surface, giving $C_2 = O(p)$, or whether some growing-prime source-valid family has two-dimensional image and $C_2 = \Theta(p^2)$.*

17 Next Experimental Checks

A useful scanner should compute, for each source-valid candidate,

$$\Delta_1 = s\tau_3 + h, \qquad G = s^2\Delta_0(\tau_1, \tau_2, -h/s),$$

then record whether $G \equiv 0$, whether $\Delta_1 \equiv 0$, the exceptional locus $s = h = 0$, the image size of

$$[\, q_1 : (p_1 - q_2) : p_2 \,],$$

and the direct slope count over distinct split triples $T \subset \mathbb{F}_p$.

The certificate should keep at least

$$p,\ q_{\text{gen}},\ q_{\text{line}},\ E,\ B_{\text{num}},\ \text{seed},\ \text{stratum},\ R_0\text{-status},$$
$$\Delta_1 \equiv 0, \quad G \equiv 0, \quad \#\{s = h = 0\}, \quad \#\,\text{image}, \quad C_2, \quad \#\{T \text{ split distinct}\}.$$

18 June 26 Normal-Form and Dependency Results

Status: mixed. This section records the parts of the June 26 PR batch that survived a statement-level validity check. The statements below are useful proof infrastructure, not prize claims. In particular, they do not improve the current Cycle116/119 numerator 52,747,567,092 over $\mathbb{F}_{17^{32}}$, and they do not prove the corrected Paper B MCA conjecture.

18.1 Syndrome-pencil normal form for support-wise line incidence

Status: New-local. This is the cleanest reusable F1 contribution from the batch. It turns support-wise line incidence into a Hankel-pencil locator equation over the actual line field.

Let $D = \{x_1, \ldots, x_n\} \subset F$ have distinct points and let $C = \mathrm{RS}[F, D, k]$ with redundancy $r = n - k$. For each x_i, put

$$\lambda_i = \left(\prod_{h \neq i}(x_i - x_h)\right)^{-1}.$$

For a word $y : D \to F$, define

$$\mathrm{Syn}(y)_m = \sum_i \lambda_i x_i^m y(x_i), \qquad 0 \leq m < r.$$

Then $\mathrm{Syn}(y) = 0$ if and only if $y \in C$. Fix $T \subset D$ with $|T| = j$, assume $0 \leq j \leq r$, and set $t = r - j$. Write

$$L_T(X) = \prod_{x \in T}(X - x) = \ell_0 + \ell_1 X + \cdots + \ell_j X^j, \qquad \boldsymbol{\ell}_T = (\ell_0, \ldots, \ell_j)^T.$$

For $w \in F^r$, define the $t \times (j+1)$ Hankel window

$$H_{t,j}(w)_{a,b} = w_{a+b}, \qquad 0 \le a < t, \quad 0 \le b \le j.$$

Theorem 4 (Hankel-pencil test)**.** *Let $f, g : D \to F$, put $u = \mathrm{Syn}(f)$ and $v = \mathrm{Syn}(g)$, and let $S = D \setminus T$. For a finite slope $z \in F$, the word $f + zg$ is explained by a codeword of C on S if and only if*

$$(H_{t,j}(u) + zH_{t,j}(v))\boldsymbol{\ell}_T = 0.$$

Moreover, this explanation is support-wise noncontained for the line $f + zg$ on S if and only if, in addition,

$$H_{t,j}(v)\boldsymbol{\ell}_T \ne 0.$$

Consequently the support-wise MCA bad slopes at agreement $k + t$ are exactly the slopes for which there is a squarefree D-split locator L_T of degree $j = r - t$ satisfying the displayed incidence and noncontainment tests.

Proof. Let $W_T \subset F^r$ be the span of the parity-check columns indexed by T:

$$W_T = \mathrm{span}_F\{\lambda_x(1, x, \ldots, x^{r-1}) : x \in T\}.$$

Equivalently, $w \in W_T$ means that w is the syndrome of a word supported on T. For every $w \in W_T$, write

$$w_m = \sum_{x \in T} c_x x^m.$$

Then, for $0 \le a < t$,

$$\left(H_{t,j}(w)\boldsymbol{\ell}_T\right)_a = \sum_{b=0}^{j} \ell_b w_{a+b} = \sum_{x \in T} c_x x^a L_T(x) = 0.$$

Conversely, since L_T is monic, the recurrence $\sum_{b=0}^{j} \ell_b w_{a+b} = 0$ for $0 \le a < r - j$ determines all remaining coordinates from the first j coordinates, so its solution space has dimension at most j. The j columns indexed by the distinct points of T form a Vandermonde system of rank j, hence the recurrence space is exactly W_T.

The word $f + zg$ is explained on S if and only if there is $c \in C$ such that $f + zg - c$ is supported on T. Taking syndromes, this is equivalent to $u + zv \in W_T$, which is the first displayed equation by the recurrence criterion. Under that equation, simultaneous explanation of f and g on the same S is equivalent to $u, v \in W_T$. Since $u = (u + zv) - zv$, this is equivalent to $v \in W_T$, hence to $H_{t,j}(v)\boldsymbol{\ell}_T = 0$. Noncontainment is therefore exactly the negation of that condition. □

18.2 Interleaved codegree reduction

Status: New-local reduction; saving remains conditional. This result is useful for the L2 interleaved-list ledger because it replaces the naive Cartesian product by a recursion through higher-agreement tails.

Let $C = \mathrm{RS}[F, D, k]$. For a word $V : D \to F$ and codeword $c \in C$, write

$$A_V(c) = \{x \in D : c(x) = V(x)\}, \qquad \mathrm{Fib}_V(s) = \{c \in C : |A_V(c)| \ge s\},$$

and $M_V(s) = |\,\mathrm{Fib}_V(s)|$. For a μ-row word $U = (U_1, \ldots, U_\mu)$, define

$$\Lambda_\mu^{(s)}(U) = \#\{(c_1, \ldots, c_\mu) \in C^\mu : |\cap_i A_{U_i}(c_i)| \ge s\}.$$

Theorem 5 (two-regime codegree bound)**.** *For every agreement threshold a,*

$$\Lambda_2^{(a)}(U_1, U_2) \leq M_{U_2}(a) + M_{U_2}(2a-k)\, M_{U_1}(a).$$

More generally,

$$\Lambda_\mu^{(a)}(U_1, \dots, U_\mu) \leq \Lambda_{\mu-1}^{(a)}(U_2, \dots, U_\mu) + \Lambda_{\mu-1}^{(2a-k)}(U_2, \dots, U_\mu)\, M_{U_1}(a).$$

Proof. For the two-row case, sum over $c_2 \in \mathrm{Fib}_{U_2}(a)$, and put $N_2 = |A_{U_2}(c_2)|$. The possible c_1's are codewords agreeing with U_1 on at least a points of $A_{U_2}(c_2)$. If $N_2 < 2a-k$, then two such c_1's would agree with each other on more than k points, forcing equality. Thus this punctured list has size at most one. If $N_2 \geq 2a-k$, use the trivial bound $M_{U_1}(a)$. Summing these two regimes gives the displayed two-row inequality.

For the μ-row case, fix $(c_2, \dots, c_\mu)$ and put

$$S = \bigcap_{i=2}^{\mu} A_{U_i}(c_i).$$

The listed c_1's must agree with U_1 on at least a points of S. If $|S| < 2a-k$, the same unique-decoding argument gives at most one such c_1. If $|S| \geq 2a-k$, use the bound $M_{U_1}(a)$. Counting the two classes of $(c_2, \dots, c_\mu)$ gives the recursion. □

Remark 5. *The reduction is unconditional. The desired L2 saving still requires an L1-family input saying that the tail lists at agreement $2a-k$ are small after quotient-periodic contributions are budgeted.*

18.3 Deep-point cap dependency split

Status: Wrapper / Audit. This result does not edit Paper D, but it explains why the headline MCA cap can be routed locally without using the imported Crites–Stewart conversion.

Proposition 6 (CS25-free route to the Paper D MCA cap)**.** *Assume the finite-field, coset, divisibility, and binomial hypotheses used in Paper D's universal cap theorem. Let $C = \mathrm{RS}[F, D, k]$, $C^+ = \mathrm{RS}[F, D, k+1]$, $q = |F|$, and suppose the elementary quotient-locator fiber construction gives a C^+-list of size*

$$L \geq \binom{N}{\rho N + 2} / |B| \geq q/k + 1$$

at the endpoint $\delta_N = 1 - \rho - 2/N$. Then there is a support-wise MCA-bad line for C with bad-slope density strictly larger than

$$\frac{1}{2k}\left(1 - \frac{n}{q}\right).$$

By support-wise MCA monotonicity, the same lower bound holds for all $\delta_N \leq \delta < 1 - \rho$.

Proof. Let $U : D \to F$ be the received word with a C^+-list $\{P_1, \dots, P_L\}$ at agreement $m = k + 2n/N$. For $\alpha \in F \setminus D$, form

$$f_\alpha(x) = \frac{U(x)}{x - \alpha}, \qquad g_\alpha(x) = -\frac{1}{x - \alpha}.$$

For every listed P_i, the slope $z = P_i(\alpha)$ gives the codeword

$$Q_i(X) = \frac{P_i(X) - P_i(\alpha)}{X - \alpha} \in F[X]_{<k}$$

on the same agreement support. Conversely every such slope comes from a listed P_i. Since $m > k$, the direction g_α cannot be explained by a degree-$< k$ polynomial on the same support, so these slopes are support-wise MCA-bad.

Averaging over $\Omega = F \setminus D$, two distinct degree-$< k+1$ polynomials collide at at most k points of Ω. Therefore some α has at least

$$M \geq \frac{L}{1 + k(L-1)/(q-n)}$$

distinct values among $P_1(\alpha), \ldots, P_L(\alpha)$. Its bad-slope density is at least

$$\frac{L(q-n)}{q(q-n+k(L-1))}.$$

This is strictly greater than $(q-n)/(2kq)$ exactly when

$$2kL > q - n + k(L-1).$$

The lower bound $L \geq q/k+1$ gives $kL \geq q+k$, so the left-minus-right quantity is at least $n+2k > 0$. This proves the endpoint bound, and monotonicity in δ extends it to the stated interval. ☐

18.4 Common code-line residual budget

Status: New-local finite MDS theorem. This M2 lemma is a guardrail for line-decoding imports: a close common code-line exception still pays for residual coordinates outside the common support.

Proposition 7 (common code-line residual budget)**.** *Let $C \subseteq F^D$ be an MDS linear code of dimension k, so every nonzero codeword has fewer than k zeros on D. Let $\ell_z = f + zg$ be a received line and suppose that for some codewords $c_f, c_g \in C$ there is a common support $S_0 \subseteq D$ of size b such that $f = c_f$ and $g = c_g$ on S_0. Fix an agreement threshold a with $a + b - n \geq k$. Put $\Omega = D \setminus S_0$,*

$$f' = f - c_f, \qquad g' = g - c_g, \qquad h = \max(1, a-b),$$

and

$$c_0 = \#\{x \in \Omega : f'(x) = g'(x) = 0\}.$$

Every support-wise noncontained slope at agreement a satisfies

$$\#\{x \in \Omega : f'(x) + zg'(x) = 0\} \geq h.$$

Consequently, if $h > c_0$, then the number of such slopes is at most

$$\left\lfloor \frac{|\Omega| - c_0}{h - c_0} \right\rfloor.$$

Proof. If ℓ_z is explained on a support S of size at least a, then

$$|S \cap S_0| \geq a - |\Omega| = a + b - n \geq k.$$

On $S \cap S_0$, the explaining codeword agrees with $c_f + zc_g$. By the MDS zero-rigidity condition, the two codewords are equal. Thus, outside S_0, the line can only be explained where $f' + zg' = 0$, and at least $h = \max(1, a-b)$ such outside zeros are needed for a noncontained support.

Each non-common residual coordinate $x \in \Omega$ with $(f'(x), g'(x)) \neq (0, 0)$ can vanish for at most one finite slope z. A bad slope needs at least $h - c_0$ private non-common residual zeros after the c_0 common zeros are counted. Charging these private coordinates to slopes gives the stated packing bound. ☐

19 Recent PR Triage: L1, F1, M1, and Challenge-Map Ledgers

Status: Audit / Experimental / bounded Proved notes. The open PR batch of June 27, 2026, adds useful proof infrastructure but no new unconditional prize-scale threshold beyond the tangent-floor consequences already in the high-agreement ledger. The promotion rule used here is conservative: bounded toy theorems and exact normal forms may enter the experimental notes, while computation-dependent numerators and generated proof programs remain labelled as such.

19.1 L1: proper-subgroup and monomial-prefix lanes

Scott Hughes' $d = 3$ proper-subgroup note records the actual H^{2k} collision object for the odd moments $1, 3, 5$. For $p > 5$, $H \le \mathbb{F}_p^*$ of order n, and

$$S_H(a_1, a_3, a_5) = \sum_{h \in H} \psi(a_1 h + a_3 h^3 + a_5 h^5),$$

the exact Fourier identity is

$$V_{H,k}^{(3)} = p^{-3} \sum_{a_1, a_3, a_5 \in \mathbb{F}_p} |S_H(a_1, a_3, a_5)|^{2k}.$$

After splitting frequencies into the linear, cubic, and quintic strata, and using the stated one-variable Gauss/Katz inputs, the note gives

$$0 \le V_{H,k}^{(3)} - \frac{n^{2k}}{p^3} \le \frac{p-1}{p^3} p^k + \frac{p(p-1)}{p^3} (3\sqrt{p})^{2k} + \frac{p^2(p-1)}{p^3} (5\sqrt{p})^{2k}.$$

This is useful L1 infrastructure, not a reserve-scale arbitrary-word locator local limit.

Vadim Avdeev's monomial-prefix note settles one bounded toy lane over

$$F = \mathbb{F}_{17}[z]/(z^{32} - 3), \qquad H = \langle z \rangle, \qquad |H| = 512,$$

with degree convention $\deg P \le 256$. In the range $258 \le A \le 512$, the admissible monomial agreement sizes are

$$258, 259, 260, 262, 264, 268, 272, 280,$$
$$288, 304, 320, 352, 384, 512.$$

The proof combines a local length-16 imbalance check over $\mathbb{F}_{17}$, a dyadic power-sum descent gate, and quotient-complement survivor families. This should be cited only as a monomial-prefix benchmark, not as an arbitrary-word list theorem.

19.2 F1: arbitrary-anchor residue clouds

The F1 arbitrary-anchor update splits the balanced residue-cloud problem from the residual-list problem. If $t < \sigma$, residue-line bad slopes inject into an ordinary list for $\mathrm{RS}[F, D, k+t]$ at residual slack $\sigma - t$. In the balanced case $t = \sigma$, arbitrary anchors have a real free-packing floor:

$$\Lambda_{\mathrm{NC}} \ge \left\lfloor \frac{|D| - k}{\sigma} \right\rfloor.$$

Moreover this is the exact ceiling for ordered k-degenerate support families, so any larger arbitrary-anchor packet must use dense overlaps. The dense-overlap gate is explicit: for two supports S, T with overlap J, compatibility of distinct slopes is controlled by whether the residue direction N/E is degree-$< k$ on J. This is a useful F1 warning and search normal form, not an extension-line lift theorem.

19.3 M1: all-line Hankel aperiodic packet

The large M1 PR is not promoted wholesale. The useful distillation is the exact Hankel-pencil object. For a line (f, g), syndromes $u = \mathrm{Syn}(f)$ and $v = \mathrm{Syn}(g)$, and a split complement T with locator vector ℓ_T, finite bad slopes are governed by

$$(H_{t,j}(u) + zH_{t,j}(v))\ell_T = 0, \qquad H_{t,j}(v)\ell_T \neq 0.$$

In the $t = 2$ window this becomes

$$b_T \neq 0, \qquad \det[a_T \; b_T] = 0, \qquad a_T = H_{2,j}(u)\ell_T, \quad b_T = H_{2,j}(v)\ell_T.$$

Same-slope one-exchange collisions lie in fixed-slope root slices, and the residual one-exchange graph can be further organized into quadratic companion slices and top packets. These reductions define a promising M1 proof program, but they do not yet prove the desired uniform polynomial bound on residual aperiodic slope images.

19.4 M1: same-slope one-exchange root slices

The first local reduction in the all-line Hankel packet is elementary and proved in `experimental/notes/m1/m1_same_slope_root_slice_lemma.md`. Let

$$L_z = H_{t,j}(u) + zH_{t,j}(v)$$

be the finite-slope landing map. Fix a $(j-1)$-core R, and write

$$T_y = R \cup \{y\}, \qquad \ell_{T_y} = (X - y)\ell_R.$$

If two distinct one-exchange extensions T_{y_1} and T_{y_2} satisfy

$$L_z\ell_{T_{y_1}} = L_z\ell_{T_{y_2}} = 0,$$

then subtracting gives

$$(y_2 - y_1)L_z\ell_R = 0.$$

Thus $L_z\ell_R = 0$, and substituting into either endpoint equation gives

$$L_z(X\ell_R) = 0.$$

Consequently

$$L_z\ell_{T_y} = 0 \qquad \text{for every } y.$$

So every same-slope one-exchange edge lies in the full fixed-slope root slice with core R. After fixed-slope root slices have been charged, the residual one-exchange graph has only different-slope edges. This is a local reduction, not a bound for the remaining different-slope codegree. The verifier `experimental/scripts/verify_m1_same_slope_root_slice_lemma.py` checks the subtraction identity and its row-wise linear-map consequence in small fields.

The same two padded equations also lift the root slice to a higher-slack Hankel core. With $w_z = u + zv$, the equations

$$H_{t,j}(w_z)(\ell_R, 0)^T = 0, \qquad H_{t,j}(w_z)(0, \ell_R)^T = 0$$

are exactly the first t and last t rows of

$$H_{t+1,j-1}(w_z)\ell_R = 0.$$

Thus a same-slope root slice is not a fresh t-level residual multiplicity: it belongs to the lifted $(t+1, j-1)$ Hankel-core ledger at the same finite slope.

The same mechanism has a two-exchange full-plane form. For a $(j-2)$-core R, write a two-root locator in elementary coordinates

$$\ell_{T_{s,p}} = (X^2 - sX + p)\ell_R, \qquad s = x + y, \quad p = xy.$$

For fixed finite slope z,

$$L_z \ell_{T_{s,p}} = L_z(X^2 \ell_R) - sL_z(X\ell_R) + pL_z(\ell_R)$$

is affine-linear in (s, p). Hence three affinely non-collinear same-slope points in the two-root plane force

$$L_z \ell_R = L_z(X\ell_R) = L_z(X^2 \ell_R) = 0,$$

equivalently

$$H_{t+2,j-2}(u + zv)\ell_R = 0.$$

Thus full two-exchange planes are charged to the lifted $(t+2, j-2)$ Hankel ledger. After fixed-root lines and full planes are removed, any residual same-slope two-exchange component through R is a line packet in the (s, p)-plane.

The full-plane statement is the $h = 2$ case of a general elementary-packet lift. For a $(j-h)$-core R, write

$$\ell_{R,c} = (X^h + c_{h-1}X^{h-1} + \cdots + c_0)\ell_R, \qquad c \in F^h.$$

For fixed finite slope z, the map $c \mapsto L_z \ell_{R,c}$ is affine-linear. Hence if $h + 1$ affinely independent coefficient points are killed at the same slope, then

$$L_z(X^m \ell_R) = 0, \qquad 0 \le m \le h.$$

These $h + 1$ padded equations are exactly the row blocks of

$$H_{t+h,j-h}(u + zv)\ell_R = 0.$$

Thus every full affine-rank same-slope h-exchange packet is charged losslessly to the lifted $(t+h, j-h)$ Hankel-core ledger. Applying the same argument separately to u and v gives the simultaneous-kernel version: full affine-rank packets in $K_{r,d}(u, v)$ lift to $K_{r+h,d-h}(u, v)$. The lower-rank case is also exact. If $C \subset F^h$ is any killed coefficient packet and $\mathrm{aff}(C) = c_* + W$, then

$$L_z(X^h \ell_R) + \sum_m c_{*,m} L_z(X^m \ell_R) = 0, \qquad \sum_m w_m L_z(X^m \ell_R) = 0 \quad (w \in W).$$

Conversely these equations kill the whole affine subpacket $c_* + W$. Thus after full-rank packets are charged, the residual same-slope elementary packets are precisely lower-rank affine subpackets; for $h = 2$ these are the lines classified next. Among coefficient hyperplanes, the fixed-root ones are exactly the evaluation hyperplanes. Namely

$$b + \sum_{m=0}^{h-1} a_m c_m = 0$$

is the locus $P_c(\alpha) = 0$ for a fixed finite root α iff, up to a nonzero scalar, $(a_0, \ldots, a_{h-1}, b) = (1, \alpha, \ldots, \alpha^h)$. Such hyperplanes are charged to the $(h-1)$-exchange root-slice ledger. For $h = 2$, this is $p-\alpha s+\alpha^2=0$, equivalently $(x-\alpha)(y-\alpha)=0$. More generally, every coefficient hyperplane has a one-root fiber dichotomy. Fix a monic $(h-1)$-root factor

$$Q_d(X)=X^{h-1}+d_{h-2}X^{h-2}+\cdots+d_0$$

and write $d_{h-1}=1, d_{-1}=0$. The extension

$$P_y=(X-y)Q_d$$

has coefficients $c_m(y)=d_{m-1}-yd_m$, so the hyperplane equation restricts to

$$b+\sum_m a_m c_m(y)=\Big(b+\sum_m a_m d_{m-1}\Big)-y\sum_m a_m d_m.$$

Thus a fixed $(h-1)$-core has at most one extension in the hyperplane unless the whole affine one-root line lies there. In the killed same-slope packet setting, such a full line is exactly a one-exchange root slice and lifts to the $(t+1,j-1)$ Hankel core. After these full one-root fibers are charged, a codimension-one coefficient-hyperplane packet has no residual one-exchange edges. The same 0/1/all dichotomy holds for any affine rank-defect packet $A=c_*+W\subset F^h$. For the same one-root fiber, put

$$d_{\mathrm{vec}}=(d_0,\ldots,d_{h-1}).$$

If two distinct roots y_1,y_2 have $c(y_i)\in A$, then

$$c(y_2)-c(y_1)=-(y_2-y_1)d_{\mathrm{vec}}\in W,$$

so $d_{\mathrm{vec}}\in W$ and the whole affine line $c(y)$ lies in A. Thus, after full one-root fibers are root-slice charged, no residual affine rank-defect packet has one-exchange edges. The two-root fiber has the same affine form. For a fixed monic $(h-2)$-root factor Q_e, write

$$P_{s,p}=(X^2-sX+p)Q_e,\qquad c_m(s,p)=e_{m-2}-se_{m-1}+pe_m.$$

If an affine rank-defect packet contains three non-collinear points of this (s,p)-plane, then both plane directions lie in W, so the whole two-root plane is contained in the packet and is charged to the $(t+2,j-2)$ full-plane lift. After full planes are charged, every residual two-root fiber intersection is at most a line packet. In general, for a moving monic r-root factor B_a multiplying a fixed $(h-r)$-root factor Q, the map $a\mapsto B_aQ$ is an affine embedding of F^r into coefficient space. The preimage of $A=c_*+W$ is therefore an affine subspace of F^r: if it has full affine rank r, the whole moving r-root fiber is contained in A and is charged to the $(t+r,j-r)$ full-packet lift; otherwise the residual intersection has dimension at most $r-1$. Over a finite field of size Q_F, this gives the formal residual count

$$\#\{a\in F^r:\mathrm{coeff}(B_aQ)\in A\}\le Q_F^{r-1}$$

on every fixed r-root fiber after the full-fiber charge; split, domain, quotient, tangent, and non-containment filters only shrink this count. Consequently the residual r-exchange graph inside one affine packet has maximum degree at most

$$\binom{h}{r}(Q_F^{r-1}-1),$$

because each vertex has only $\binom{h}{r}$ fixed $(h-r)$-subfactors to complete.

The residual two-root lines are exactly the expected models: if

$$As + Bp + C = 0$$

and $B = 0$, then $x + y = s_0$ is a fixed-sum packet; if $B \neq 0$, then after writing $c = -A/B$, $\beta = -C/B$, the equation becomes

$$(x - c)(y - c) = c^2 + \beta.$$

The zero right-hand side is a fixed-root line, and the nonzero right-hand side is a product-Mobius packet. Hence after fixed-root lines are charged, the only residual line packets are fixed-sum and nondegenerate product-Mobius packets.

In the actual $t = 2$ Hankel window, these non-fixed line packets cannot be constant-slope residuals. If

$$H_{2,j}(u + zv)\ell_{R,s,p} = (d_2 - sd_1 + pd_0,\ d_3 - sd_2 + pd_1)$$

vanishes identically on a fixed-sum line $s = s_0$, then the coefficient of p gives $(d_0, d_1) = 0$, and the Hankel overlap forces $d_2 = d_3 = 0$. If it vanishes identically on a nondegenerate product-Mobius line $p = cs - c^2 + \mu$, $\mu \neq 0$, then $(d_1, d_2) = c(d_0, d_1)$ and $(d_2, d_3) = (c^2 - \mu)(d_0, d_1)$, so $\mu d_0 = 0$ and again all four d_i vanish. Thus every constant-slope non-fixed line packet is already charged to the full-plane lift $H_{4,j-2}(u + zv)\ell_R = 0$. After fixed-root and full-plane charges, the active finite-slope map on each surviving fixed-sum or product-Mobius packet is injective.

In the $t = 2$ Hankel window the same one-exchange core has a quadratic determinant gate. Put

$$a_y = H(u)\ell_{T_y} = a_X - ya_0, \qquad b_y = H(v)\ell_{T_y} = b_X - yb_0 \qquad \text{in } F^2.$$

Then the finite-slope determinant condition is

$$\Delta_R(y) = \det(a_y, b_y) = 0, \qquad b_y \neq 0,$$

where

$$\Delta_R(y) = \det(a_X, b_X) - y\big(\det(a_0, b_X) + \det(a_X, b_0)\big) + y^2 \det(a_0, b_0).$$

Thus a nonzero Δ_R admits at most two anchors y. If three distinct anchors pass the determinant gate, then $\Delta_R \equiv 0$, so the core is a ruled determinant branch. The ruled branch first has the abstract affine dichotomy: if

$$\det(a_X - ya_0,\ b_X - yb_0) \equiv 0,$$

then either there is a fixed finite slope z_0 with

$$a_y + z_0 b_y = 0 \qquad \text{for every } y,$$

or $b_y = 0$ for every y, or all four vectors

$$a_X,\ a_0,\ b_X,\ b_0$$

lie in a single output line. In the last case write

$$a_y = \alpha(y)c, \qquad b_y = \beta(y)c.$$

For arbitrary affine pencils this last case could be a moving-slope branch, but the Hankel shift structure rules it out here. If

$$c_i(w) = \mathrm{row}_i(H_{3,j-1}(w)\ell_R), \qquad i = 0, 1, 2,$$

then

$$H_{2,j}(w)\ell_{T_y} = (c_1(w) - yc_0(w),\ c_2(w) - yc_1(w)).$$

A common output line imposes two recurrence equations on $(c_0(w), c_1(w), c_2(w))$, whose solution space is one-dimensional. Applying this to $w = u$ and $w = v$ makes the two Hankel pencils proportional, unless $b_y = 0$ identically. Therefore every active ruled Hankel core is already a fixed finite-slope core; after fixed-slope root slices are charged, ruled cores contribute no residual one-exchange edges.

The same note also records the exact one-exchange triangle dichotomy in the Johnson graph. Three j-set locators that are pairwise one-exchange are either a star

$$T_i = R \cup \{y_i\}, \qquad |R| = j - 1,$$

or a top-packet triangle

$$T_i = U \setminus \{x_i\}, \qquad |U| = j + 1.$$

Thus every active star triangle is precisely a three-anchor event through one $(j-1)$-core, hence a ruled determinant core by the quadratic gate above. Since ruled Hankel cores collapse to fixed-slope or inactive cores, residual one-exchange triangles after fixed-slope root-slice charging are therefore top-packet triangles.

Top packets then lift losslessly to a $t = 1$ Hankel kernel. If

$$T_x = U \setminus \{x\}, \qquad \ell_U = (X - x)\ell_{T_x},$$

then for every syndrome vector w,

$$H_{1,j+1}(w)\ell_U = \mathrm{row}_1\big(H_{2,j}(w)\ell_{T_x}\big) - x\,\mathrm{row}_0\big(H_{2,j}(w)\ell_{T_x}\big).$$

Thus a $t = 2$ incidence for T_x at slope z implies

$$(H_{1,j+1}(u) + zH_{1,j+1}(v))\ell_U = 0.$$

If two members of the same top packet have distinct slopes, subtracting the two lifted scalar equations gives

$$H_{1,j+1}(u)\ell_U = H_{1,j+1}(v)\ell_U = 0.$$

Therefore every residual top-packet edge after fixed-slope charging lies over a common lifted $t = 1$ kernel. This turns the top-packet triangle residual into a higher-slack kernel target rather than a fresh one-exchange multiplicity phenomenon.

Writing

$$K_{\mathrm{top}}(u,v) = \{\, U \subset D :\ |U| = j+1,\ H_{1,j+1}(u)\ell_U = H_{1,j+1}(v)\ell_U = 0 \,\},$$

the top-packet branch has the explicit compression ledger

$$E_{\mathrm{top}}(A_{\mathrm{res}}) \le \binom{j+1}{2} |K_{\mathrm{top}}(u,v)|, \qquad \mathrm{Tri}_1(A_{\mathrm{res}}) \le \binom{j+1}{3} |K_{\mathrm{top}}(u,v)|.$$

Indeed a residual top edge recovers its union U, and (because residual one-exchange edges have distinct slopes) the lift above forces $U \in K_{\mathrm{top}}(u,v)$. Inside such a packet,

$$H_{2,j}(w)\ell_{U\setminus\{x\}} = \rho_x(w)(1, x), \qquad w \in \{u,v\},$$

so the remaining slope label is the scalar ratio $\rho_x(u) + z\rho_x(v) = 0$, $\rho_x(v) \neq 0$. Thus top-packet edges and triangles are charged to the lifted $t = 1$ top-kernel family.

The top-kernel family also has an exact root-slice recursion. More generally, for $r \geq 1$ and locator size d, put

$$K_{r,d}(u,v) = \{\, U \subset D : \ |U| = d, \ H_{r,d}(u)\ell_U = H_{r,d}(v)\ell_U = 0 \,\}.$$

If $U_{y_i} = R \cup \{y_i\} \in K_{r,d}(u,v)$ for two distinct anchors through the same $(d-1)$-core R, then the same subtraction argument applied separately to u and v gives

$$H_{r,d}(w)\ell_R = H_{r,d}(w)(X\ell_R) = 0, \qquad w \in \{u,v\}.$$

These are exactly the first r and last r rows of

$$H_{r+1,d-1}(w)\ell_R = 0.$$

Hence $R \in K_{r+1,d-1}(u,v)$. So one-exchange collisions inside the lifted top-kernel ledger are charged losslessly to the next simultaneous Hankel kernel; after those root slices are paid, the residual $K_{r,d}$ family has no one-exchange edges. This records the residual-depth frontier shift as an identity rather than a multiplicative loss source.

The same Hankel shift also classifies the one-outside external-anchor ruled branch from the boundary-off audit. For a shadow $S \subset D$, $|S| = j - 1$, and an external anchor $\beta \notin D$,

$$\ell_{S,\beta} = (X - \beta)\ell_S.$$

The $t = 2$ determinant gate is again quadratic in β. If it is nonzero, at most two external anchors pass. If it vanishes identically, then with

$$c_i(w) = \mathrm{row}_i(H_{3,j-1}(w)\ell_S)$$

one has

$$H_{2,j}(w)\ell_{S,\beta} = (c_1(w) - \beta c_0(w), \ c_2(w) - \beta c_1(w)).$$

The same recurrence argument as above forces the ruled external-anchor branch to be inactive or fixed finite slope. In the fixed-slope case it is a boundary root slice and lifts to

$$H_{3,j-1}(u + z_0 v)\ell_S = 0.$$

Thus, after fixed-slope boundary slices are charged, each shadow S supports at most two active non-ruled external anchors. This classifies the ruled $\mathrm{Boundary}_{\mathrm{off}}$ branch locally, but it is not yet a global bound for the one-outside target image. Equivalently, projecting a residual one-outside target $B_\beta = S \cup \{\beta\}$ to its domain shadow S has fibers of size at most two. If two external anchors over the same shadow had the same finite slope z, then subtraction gives the lifted boundary root slice

$$H_{3,j-1}(u + zv)\ell_S = 0,$$

so after that charge the remaining two-anchor shadow fibers have distinct finite slopes. Hence

$$|\mathrm{Boundary}_{\mathrm{off}}^{\mathrm{res}}| \leq 2|\mathrm{Shadow}_{\mathrm{off}}^{\mathrm{res}}|.$$

This reduces the live one-outside object to a shadow-image problem rather than an external-anchor multiplicity problem. The shadow condition itself is intrinsic. Writing

$$c_i = \mathrm{row}_i(H_{3,j-1}(u + zv)\ell_S), \qquad 0 \leq i \leq 2,$$

there is an external anchor β with $H_{2,j}(u+zv)\ell_{S,\beta}=0$ iff either $(c_0,c_1,c_2)=(0,0,0)$, the already charged lifted boundary slice, or

$$c_0\neq 0,\qquad c_1^2=c_0c_2,\qquad \beta=c_1/c_0.$$

Thus the residual boundary shadow image lies in a rank-one scalar Hankel locus with recovered anchor $c_1/c_0\notin D$, plus the usual active filter. As z varies, write $c_i(z)=a_i+zb_i$. The recovered-anchor gate is the quadratic

$$Q_S(z)=c_1(z)^2-c_0(z)c_2(z).$$

If Q_S is nonzero, the fixed shadow contributes at most two candidate finite slopes and hence at most two recovered anchors. If $Q_S\equiv 0$, the line $(c_0,c_1,c_2)(z)=a+zb$ lies on one generator of the rank-one cone $x_1^2=x_0x_2$. Thus any finite recovered anchor is independent of z, and both $H_{2,j}(u)\ell_{S,\beta}$ and $H_{2,j}(v)\ell_{S,\beta}$ vanish there. The active filter removes this degenerate branch, while zero triples are already charged to the lifted boundary slice. The remaining boundary-shadow problem is therefore a nonzero quadratic-root problem over the shadows. Equivalently, with $r_\beta=(1,\beta,\beta^2)$ and $h_\beta(c)=(c_1-\beta c_0,c_2-\beta c_1)$, define the anchor gate

$$A_S(\beta)=\det(a,b,r_\beta).$$

Then $A_S(\beta)=\det(h_\beta(a),h_\beta(b))$. If $h_\beta(b)\neq 0$, a root $A_S(\beta)=0$ gives a unique finite slope z with $h_\beta(a+zb)=0$; nonzero triples are exactly the recovered rank-one anchors above, while zero triples are charged. If $A_S\equiv 0$, then a,b are proportional and the branch is inactive or charged. Thus the same residual target is a nondegenerate conic-secant incidence: a nonzero quadratic in the outside anchor β, with the active filter imposed. Fixing β gives a further one-root reduction. For $|R|=j-2$, put $P_{R,\beta}=(X-\beta)\ell_R$ and $\ell_{R,\beta,y}=(X-y)P_{R,\beta}$. The determinant

$$\det(H_{2,j}(u)\ell_{R,\beta,y},H_{2,j}(v)\ell_{R,\beta,y})$$

is quadratic in y. A nonzero determinant has at most two domain roots; an identically zero determinant is inactive or fixed finite slope, and the fixed-slope case lifts to

$$H_{3,j-1}(u+zv)P_{R,\beta}=0.$$

After charging these one-outside lifted boundary-core slices, projection $(R\cup\{y\},\beta)\mapsto(R,\beta)$ has residual fibers of size at most two. Combining this with the shadow-fiber bound, for each fixed $(j-2)$-core R the residual boundary incidences between external anchors β and domain roots $y\in D\setminus R$ form a bipartite graph of degree at most two on both sides. Hence each fixed R supports at most $2(n-j+2)$ residual external anchors. Thus the next boundary object is the lower-dimensional boundary-core image, or still further the active $(j-2)$-core image. For fixed R, write

$$U_m=H_{2,j}(u)(X^m\ell_R),\qquad V_m=H_{2,j}(v)(X^m\ell_R),\qquad 0\le m\le 2.$$

Then the remaining fixed-core incidence is cut out by the symmetric bidegree $(2,2)$ determinant

$$\Delta_R(\beta,y)=\det(U_2-(\beta+y)U_1+\beta yU_0,\ V_2-(\beta+y)V_1+\beta yV_0)\,.$$

The fixed-shadow and fixed-anchor quadratics are the two coordinate specializations of this same determinant. Thus the active core image is a concrete bidegree-two incidence target, not an uncontrolled external-anchor search. Equivalently, $\Delta_R(\beta,y)=F_R(\beta+y,\beta y)$, where

$$F_R(s,p)=\det(U_2-sU_1+pU_0,\ V_2-sV_1+pV_0)$$

is the ordinary two-root elementary determinant. Fixing either coordinate is the fixed-root line $p = \alpha s - \alpha^2$ in the (s, p)-plane, so the boundary-core target is a mixed-domain slice of the same determinant geometry used by the fixed-root, fixed-sum, and product-Mobius line-packet ledger. Moreover the mixed slice of each surviving non-fixed line packet is explicit: on a fixed-sum line $x + y = s_0$ one has $\beta = s_0 - y$, while on a nondegenerate product-Mobius line $(x - c)(y - c) = \mu \neq 0$ one has $\beta = c + \mu/(y - c)$. Thus, over a fixed core R, such a packet contributes at most $n - j + 2$ one-outside boundary-core pairs, namely the domain roots whose packet partner leaves D; the $\mu = 0$ case is the fixed-root line already charged. The same elementary-plane form also controls slope multiplicity on the remaining conic branch. For fixed finite slope z, put $W_m = U_m + zV_m$. The same-slope equations are

$$W_2 - sW_1 + pW_0 = 0,$$

a pair of affine-linear equations in (s, p). Hence the same-slope fiber is empty, a point, an affine line, or the full plane. A repeated slope on two distinct elementary points therefore forces a constant-slope affine line, or the full-plane lift. Fixed-root lines are already boundary/root-slice charges, and non-fixed constant-slope lines collapse to the full-plane charge above. Thus after these charges the finite-slope label is injective on the live fixed-core boundary conic, so the quartic point count below also controls the number of live slopes. For the complementary conic case, write

$$F_R(s, p) = f_{20}s^2 + f_{11}sp + f_{02}p^2 + f_{10}s + f_{01}p + f_{00}.$$

Then for fixed y,

$$F_R(\beta + y, \beta y) = A_R(y)\beta^2 + B_R(y)\beta + C_R(y),$$

where

$$\begin{aligned}
A_R(y) &= f_{20} + f_{11}y + f_{02}y^2, \\
B_R(y) &= f_{10} + (2f_{20} + f_{01})y + f_{11}y^2, \\
C_R(y) &= f_{00} + f_{10}y + f_{20}y^2.
\end{aligned}$$

In odd characteristic the nondegenerate anchor count is controlled by the quartic discriminant $B_R(y)^2 - 4A_R(y)C_R(y)$; a fully zero quadratic is the fixed-root component $p = ys - y^2$, already charged. Thus the remaining non-line boundary-core target is a quartic discriminant/Kummer trace over domain roots, before outside-domain and active filters. If this quartic discriminant is identically zero, then coefficient comparison shows that either $F_R = \lambda L^2$ for an affine line L, already a fixed-sum, fixed-root, or product-Mobius line-packet branch, or $F_R = \lambda(s^2 - 4p)$. The latter restricts to $\lambda(\beta - y)^2$ on the boundary slice, hence contributes only $\beta = y$, which is removed by the outside-domain/distinct-root filters. So the live conic target after line-packet and fixed-root charges has nonzero quartic discriminant. If $A_R \not\equiv 0$, then away from at most two roots of A_R the quadratic formula identifies anchor roots with the double cover

$$W_R^2 = B_R(y)^2 - 4A_R(y)C_R(y), \qquad W_R = 2A_R(y)\beta + B_R(y).$$

After squarefree reduction this cover has genus at most one, since the discriminant has degree at most four. If $A_R \equiv 0$, the determinant is affine-linear in (s, p) and has already returned to the line-packet branch. Thus the live non-line boundary target has uniformly bounded genus/conductor, up to at most two linear exceptional fibers. There is also an exact full-subgroup Kummer gate. If $D \leq \mathbb{F}_p^*$ has index e, χ has kernel D, and $\ell = \mathrm{lcm}(e, 2)$, then away from the charged fixed-root fibers and the at most two $A_R = 0$ exceptions the full-subgroup count expands as

$$N_R(D) = |D| + \frac{1}{e} \sum_{a=0}^{e-1} \sum_{y \in \mathbb{F}_p^*} \chi^a(y)\chi_2(\mathrm{Disc}_R(y)) + O(1).$$

Combining χ and χ_2 through a character of order ℓ, the a-term is geometrically trivial only if

$$\operatorname{div}\left(y^{a\ell/e}\,\operatorname{Disc}_R(y)^{\ell/2}\right) \equiv 0 \pmod{\ell}.$$

Equivalently, all nonzero roots of Disc_R have even multiplicity and the zero-root parity matches a. Hence the squarefree part of a no-cancellation quartic is forced to be either 1 or y:

$$\operatorname{Disc}_R(y) = cG(y)^2 \qquad \text{or} \qquad \operatorname{Disc}_R(y) = c\,y\,G(y)^2.$$

The first case forces $a = 0$, while the second forces e even and $a = e/2$. If this power gate fails, the standard one-dimensional Kummer-Weil bound on $\mathbb{P}^1$ has support contained in the four discriminant roots plus 0 and ∞, so the term has depth-independent conductor and is bounded by $4\sqrt{p}$. Thus any no-cancellation boundary quartic is an explicit rational graph or fixed square-root rational graph branch, not a hidden growing-conductor family. It follows that the full-subgroup live count over a fixed core satisfies

$$N_R^{\text{live}}(D) \le |D| + 4\sqrt{p} + O(1)$$

when the power gate is absent, and

$$N_R^{\text{live}}(D) \le 2|D| + 4\sqrt{p} + O(1)$$

in general. There is at most one geometrically trivial Fourier term: the principal term in the cG^2 case or the quadratic subgroup term in the cyG^2 case. The outside-domain and active filters only delete points, and the same-slope fiber injection above turns this point count into the same live slope count for the fixed core. Combining with the degree-two graph bound gives the ledger-facing non-line conic estimate

$$N_R^{\text{conic}}(D) \le \min\{2(n-j+2),\, |D| + 4\sqrt{p} + O(1)\}$$

when the power gate is absent, and

$$N_R^{\text{conic}}(D) \le \min\{2(n-j+2),\, 2|D| + 4\sqrt{p} + O(1)\}$$

in general. Reducible fixed-sum and product-Mobius line packets remain in the separate graph/packet ledger; the non-line fixed-core geometry has no remaining uncontrolled multiplicity. The same fixed-core target also has a slope-side recurrence chart. Write

$$a_i = \operatorname{row}_i(H_{4,j-2}(u)\ell_R), \qquad b_i = \operatorname{row}_i(H_{4,j-2}(v)\ell_R), \qquad c_i(z) = a_i + zb_i.$$

For $P(X) = X^2 - sX + p$, the equation

$$H_{2,j}(u+zv)(P\ell_R) = 0$$

is exactly

$$c_2 - sc_1 + pc_0 = 0, \qquad c_3 - sc_2 + pc_1 = 0.$$

If $Q_R(z) = c_0c_2 - c_1^2 \neq 0$, then

$$s = \frac{c_0c_3 - c_1c_2}{Q_R(z)}, \qquad p = \frac{c_1c_3 - c_2^2}{Q_R(z)},$$

and the split-root condition is governed by the quartic numerator

$$(c_0c_3 - c_1c_2)^2 - 4(c_0c_2 - c_1^2)(c_1c_3 - c_2^2).$$

If $Q_R(z) = 0$ and a solution exists, then the four c_i's are either all zero, giving the full-plane lift, or form a geometric progression $c_i = c_0\alpha^i$, in which case every solution has $P(\alpha) = 0$ and is a fixed-root line. Thus the residual non-line root-core target lives in the nonzero-denominator recurrence chart. Set

$$A_R = c_0c_3 - c_1c_2, \qquad B_R = c_1c_3 - c_2^2, \qquad \Theta_R = A_R^2 - 4Q_RB_R.$$

Because the c_i's are linear in z, the polynomials Q_R, A_R, B_R have degree at most two and Θ_R has degree at most four. On $Q_R \neq 0$, a split lift with roots r_1, r_2 gives

$$y = Q_R(z)(r_1 - r_2), \qquad y^2 = \Theta_R(z),$$

and conversely a point on $Y^2 = \Theta_R(z)$ recovers the formal roots $(s_R(z) \pm y/Q_R(z))/2$. Therefore, after the denominator-zero fixed-root/full-plane cases are charged, the residual split root-core slopes are filtered points on a degree-four double cover of the slope line. Its squarefree reduction has genus at most one; if Θ_R is a square the roots are rational functions of z, and otherwise the standard Kummer-Weil bound has depth-independent conductor. The domain/outside filter does not change this conductor scale. If $D = (\mathbb{F}_p^*)^e$ and χ is the quotient character of order e, then on the cover, away from $Q_R = 0$,

$$r_\pm(z, Y) = \frac{A_R(z) \pm Y}{2Q_R(z)}, \qquad 1_D(x) = \frac{1}{e}\sum_{a=0}^{e-1} \chi^a(x),$$

with characters extended by zero at $x = 0$. Hence the condition $r_+ \in D$, $r_- \notin D$ is an explicit finite sum of Kummer traces

$$\sum_{(z,Y)} \chi^a(r_+(z, Y))\chi^b(r_-(z, Y))$$

on the same genus-zero/genus-one cover. The finite poles lie over the at most two roots of Q_R; finite zeros lie over the at most two roots of B_R; the remaining support is at infinity and at the at-most-four branch points of Θ_R. Thus every non-power summand has $O_e(\sqrt{p})$ size by the standard curve Kummer-Weil bound with conductor independent of depth. The only no-cancellation cases are explicit power-divisor congruences for $r_+^a r_-^b$ on this bounded divisor support. If $B_R \equiv 0$, then $p_R = 0$ on $Q_R \neq 0$, so every formal two-root extension has the fixed root 0; this is the fixed-root line $p = 0$ and is already charged. Otherwise the zero-root points lie over the at most two roots of B_R. On the open cover $C_R^\times$, where $Q_R r_+ r_- \neq 0$, put

$$N_R(z) = B_R(z)/Q_R(z) = r_+(z, Y)r_-(z, Y).$$

Then

$$1_D(r_+)(1 - 1_D(r_-)) = 1_D(r_+)(1 - 1_D(N_R(z))).$$

Indeed, once $r_+ \in D$, the condition $r_- \in D$ is equivalent to $N_R(z) \in D$. Thus the outside-root test is a slope-line norm filter; the remaining cover-level condition is only $r_+ \in D$. Equivalently,

$$N_R^{+,-}(D) = \frac{1}{e}\sum_a \sum_{C_R^\times} \chi^a(r_+)(1 - 1_D(N_R(z))) + O(1),$$

which is the same as the expansion below after reindexing $S_{a+b,b} = \sum \chi^a(r_+)\chi^b(N_R)$. For a fixed slope with $Q_RB_R \neq 0$, the mixed ordered points inject into the norm-outside slope set: if $N_R(z) \in$

D, then no mixed point occurs, while if $N_R(z) \notin D$, at most one of the two roots can lie in D. Thus

$$N_R^{+,-}(D) \le Z_R^{\mathrm{norm-out}} := \sum_{Q_R B_R \neq 0} (1 - 1_D(B_R/Q_R)) \,.$$

The right side has the $\mathbb{P}^1$ expansion

$$Z_R^{\mathrm{norm-out}} = (1 - 1/e)\#\{Q_R B_R \neq 0\} - \frac{1}{e} \sum_{a=1}^{e-1} \sum_{Q_R B_R \neq 0} \chi^a(B_R/Q_R).$$

Hence, unless B_R/Q_R is a quotient-character power, the standard $\mathbb{P}^1$ Kummer bound gives

$$N_R^{+,-}(D) \le (e-1)|D| + C_e\sqrt{p} + O_e(1).$$

The norm-power exception is explicit: for a nonprincipal χ^a of order $m = e/\gcd(e, a)$, geometric triviality is exactly $\mathrm{div}(B_R/Q_R) \equiv 0 \pmod{m}$. Since B_R, Q_R have degree at most two, $m \ge 3$ forces B_R/Q_R to be constant, while $m = 2$ says that the reduced rational function B_R/Q_R is a square up to scalar. Thus odd index has only the constant-norm exception, and even index has only the additional quadratic square-norm branch. The constant-norm branch is already charged: $B_R/Q_R = \gamma$ means $p_R(z) = \gamma$, hence every live two-root extension lies on the elementary line $p = \gamma$, i.e. the fixed-root line if $\gamma = 0$ and otherwise the center-zero product-Mobius packet $xy = \gamma$. In the remaining nonconstant square-norm case, say $B_R/Q_R = \gamma M(z)^2$ and e is even, all nonprincipal terms in the norm-outside expansion are bounded-support $\mathbb{P}^1$ Kummer sums except the quadratic coefficient $\chi^{e/2}$, where $\chi^{e/2}(B_R/Q_R) = \chi^{e/2}(\gamma) = \epsilon$. Therefore

$$Z_R^{\mathrm{norm-out}} = \frac{e-1-\epsilon}{e}\#\{Q_R B_R \neq 0\} + O_e(\sqrt{p}) + O_e(1).$$

Thus the positive-parity square-norm branch has the genus-free bound $(e-2)|D| + C_e\sqrt{p} + O_e(1)$, while the negative-parity square-norm branch has the full slope-line main term and remains the square-norm obstruction for the cover-level sums. In the positive-parity branch, writing $N_R = \gamma M^2$, the condition $N_R \in D$ is $\chi(M)^2 = \chi(\gamma)^{-1}$. This cuts out two quotient classes of the degree-one slope parameter M, so the norm-outside slopes are exactly the remaining $e-2$ classes, up to zeros and poles. In the negative-parity branch $N_R \notin D$ on the open slope line, so the mixed count collapses to

$$N_R^{+,-}(D) = e^{-1}|C_R^\times| + e^{-1}\sum_{a=1}^{e-1} S_a^+ + O(1).$$

All nonquadratic one-root terms are then nontrivial; the only possible large term is the explicit branch where r_+ is a square. Since $\deg r_+ \le 2$, this branch is fixed-root or rational. Thus the negative-parity square-norm genus-one branch has bound $|D| + C_e\sqrt{p} + O_e(1)$, while the rational one-root-square branch has a final sign: if $r_+ = \alpha H^2$, then $\chi^{e/2}(\alpha) = -1$ cancels the main term and $\chi^{e/2}(\alpha) = +1$ gives the safe bound $2|D| + C_e\sqrt{p} + O_e(1)$. In the positive-sign case the degree-one parameter $h = H$ satisfies $r_+(h) = \alpha h^2$, so $r_+(h) \in D$ is exactly the two-coset condition $\chi(h)^2 = \chi(\alpha)^{-1}$. These degree-one square-map packets have no hidden high-overlap energy. If M_1, M_2 are nonconstant degree-one rational parameters and $\chi^a(M_1)\chi^b(M_2)$ is geometrically trivial for $0 < a, b < e$, then either

$$M_2 = \lambda M_1, \qquad a + b \equiv 0 \pmod{e},$$

or

$$M_2 = \lambda/M_1, \qquad a - b \equiv 0 \pmod{e}.$$

Indeed, otherwise one zero or pole appears in only one of the two degree-one divisors and has nonzero valuation modulo e. Hence two square-map packets intersect with the expected product density $4e^{-2}p + O_e(\sqrt{p})$ unless their parameters are quotient-parallel or quotient-inverse; in the exceptional cases the intersection is again a single explicit quotient-class packet and is bounded by $2|D| + O_e(1)$. Equivalently, high-overlap square-map packets are grouped by their unordered zero-pole support $\Pi = \{\mathrm{zero}(M), \mathrm{pole}(M)\} \subset \mathbb{P}^1$. For fixed $\Pi = \{\alpha, \beta\}$, choose M_Π with divisor $\alpha - \beta$. Every degree-one parameter with this support is λM_Π or λ/M_Π, and all its two-coset packets are among

$$P_\Pi(\theta) = \{z : \chi(M_\Pi(z))^2 = \theta\}, \qquad \theta \in (\mathbb{Z}/e\mathbb{Z})^2.$$

There are exactly $e/2$ such packets; they are disjoint on $\mathbb{P}^1 \setminus \Pi$ and partition that open line. Thus a quotient-parallel/inverse high-overlap component contributes only an $O_e(1)$ finite palette of distinct slope sets per zero-pole support. In the square-norm branch the support itself is determined by the norm boundary. If $N_R = B_R/Q_R = \gamma M^2$ with M nonconstant of degree one, then

$$\mathrm{div}(N_R) = 2\,\mathrm{div}(M).$$

After reducing B_R/Q_R, the numerator and denominator are scalar multiples of squares of linear forms on $\mathbb{P}^1$, and their roots are exactly the zero and pole endpoints of $\Pi(M)$. Finite zero endpoints are roots of B_R, finite pole endpoints are roots of Q_R, and infinity accounts for degree imbalance. Thus the square-norm palette is indexed by the bounded norm-boundary divisor of the root-core recurrence chart; changing the square root only rescales or inverts M, so it stays in the same finite palette. Since B_R and Q_R have degree at most two, this square-norm branch is also a repeated-endpoint gate. After cancelling the gcd of B_R and Q_R, every finite zero or pole of the reduced norm has valuation ± 2, and the point at infinity has valuation 0 or ± 2. Equivalently, each nonconstant reduced quadratic factor has discriminant zero and reduced degree-one factors cannot occur. Thus the square-norm packet image lies on the repeated-root/discriminant-zero norm-boundary locus, rather than on the full generic root image of B_R and Q_R. Equivalently, every finite endpoint has a derivative certificate: with $\bar{B}_R, \bar{Q}_R$ the reduced numerator and denominator, a zero endpoint satisfies $\bar{B}_R(z_0) = \bar{B}'_R(z_0) = 0$ and $\bar{Q}_R(z_0) \neq 0$, while a pole endpoint satisfies $\bar{Q}_R(z_0) = \bar{Q}'_R(z_0) = 0$ and $\bar{B}_R(z_0) \neq 0$. Hence the exceptional support image is a double-root/discriminant-zero projection as the root core varies, not a generic moving-quadratic root image. For finite square-norm endpoints the bars can be removed: a common finite factor of B_R and Q_R would leave valuation of absolute value at most one after cancellation, impossible for a square-norm endpoint. Thus finite zero endpoints satisfy $B_R = B'_R = 0$ and $Q_R \neq 0$, while finite pole endpoints satisfy $Q_R = Q'_R = 0$ and $B_R \neq 0$. Writing $B_R = b_0 + b_1 z + b_2 z^2$ and $Q_R = q_0 + q_1 z + q_2 z^2$, this means a live finite endpoint forces $b_1^2 - 4b_0 b_2 = 0$ or $q_1^2 - 4q_0 q_2 = 0$; when the quadratic coefficient is nonzero the endpoint slope is $-b_1/(2b_2)$ or $-q_1/(2q_2)$. Hence moving square-norm supports are controlled by raw quadratic discriminants. In the recurrence rows $c_i(z) = a_i + z b_i$, these quadratics are the adjacent Hankel minors

$$H_i(z) = c_i(z) c_{i+2}(z) - c_{i+1}(z)^2,$$

with coefficients

$$h_{i,0} = a_i a_{i+2} - a_{i+1}^2, \quad h_{i,1} = a_i b_{i+2} + a_{i+2} b_i - 2a_{i+1} b_{i+1}, \quad h_{i,2} = b_i b_{i+2} - b_{i+1}^2.$$

Thus pole endpoints force the discriminant of $H_0 = Q_R$ to vanish, and zero endpoints force the discriminant of $H_1 = B_R$ to vanish. Equivalently, if

$$P_i = a_i b_{i+1} - a_{i+1} b_i, \quad R_i = a_{i+1} b_{i+2} - a_{i+2} b_{i+1}, \quad S_i = a_i b_{i+2} - a_{i+2} b_i,$$

then $\mathrm{disc}(H_i) = S_i^2 - 4P_iR_i$. Endpoint production therefore forces the adjacent row-pair Plucker minors onto the conic $S_i^2 = 4P_iR_i$, with the proportional-row case as the rank-one degeneracy. On the chart $P_i \neq 0$, this conic is parametrized by $\lambda_i = S_i/(2P_i)$, so $S_i = 2P_i\lambda_i$ and $R_i = P_i\lambda_i^2$. On the complementary chart $P_i = 0$, one has $S_i = 0$; if $R_i \neq 0$, then the first row-pair is zero, and symmetrically if $R_i = 0$ and $P_i \neq 0$, then the third row-pair is zero. Thus the endpoint-support ledger splits into zero-row/proportional degeneracies and the residual moving Plucker chart. In row form, if $r_j = (a_j, b_j)$ and $P_i \neq 0$, then r_i, r_{i+1} are a basis and the chart is equivalent to

$$r_{i+2} = 2\lambda_i r_{i+1} - \lambda_i^2 r_i, \qquad \lambda_i = \frac{S_i}{2P_i}.$$

The reverse chart $R_i \neq 0$ gives the symmetric formula $r_i = 2\mu_i r_{i+1} - \mu_i^2 r_{i+2}$, $\mu_i = S_i/(2R_i)$. Thus, after the degenerate charges are removed, the residual endpoint-support row triple is a one-parameter recurrence rather than a generic conic in row space. Equivalently, on the $P_i \neq 0$ chart the adjacent Hankel minor itself factors as

$$H_i = c_i c_{i+2} - c_{i+1}^2 = -(c_{i+1} - \lambda_i c_i)^2.$$

On the reverse chart it is

$$H_i = -(c_{i+1} - \mu_i c_{i+2})^2.$$

Thus the double endpoint is the zero of an explicit row-linear form, with the constant-linear case accounting for the projective endpoint at infinity. In particular, when $b_{i+1} - \lambda_i b_i \neq 0$, the finite endpoint is

$$z_i = \frac{\lambda_i a_i - a_{i+1}}{b_{i+1} - \lambda_i b_i},$$

and when this denominator vanishes the numerator is nonzero because $P_i \neq 0$, so the double endpoint is at infinity. The reverse chart gives

$$z_i = \frac{\mu_i a_{i+2} - a_{i+1}}{b_{i+1} - \mu_i b_{i+2}}.$$

Thus a fixed adjacent row basis sends the Plucker chart parameter to endpoint slope by a projective line map, not by an arbitrary quadratic root map. The adjacent zero and pole endpoint conditions also overlap rigidly. If the two Plucker conics for $i = 0, 1$ both hold and $P_0 \neq 0$, then with $\lambda_0 = S_0/(2P_0)$,

$$r_2 = 2\lambda_0 r_1 - \lambda_0^2 r_0, \qquad P_1 = R_0 = P_0\lambda_0^2.$$

If $\lambda_0 = 0$, this is the proportional-row degeneracy for the second triple. If $\lambda_0 \neq 0$, then $P_1 \neq 0$ and the second chart gives

$$r_3 = 2\lambda_1 r_2 - \lambda_1^2 r_1 = -2\lambda_1\lambda_0^2 r_0 + (4\lambda_0\lambda_1 - \lambda_1^2) r_1.$$

Thus simultaneous finite endpoint production is either charged already or is contained in an explicit two-step row-recurrence packet. In the finite-finite nondegenerate case this packet is recovered from its ordered endpoint pair. Since $P_0 \neq 0$, c_0 and c_1 have no common finite zero, so an H_0 endpoint z_0 gives

$$\lambda_0 = \frac{c_1(z_0)}{c_0(z_0)}.$$

Then an H_1 endpoint z_1 has $c_1(z_1) \neq 0$ and gives

$$\lambda_1 = 2\lambda_0 - \lambda_0^2 \frac{c_0(z_1)}{c_1(z_1)}.$$

Thus, for fixed initial row basis r_0, r_1, an ordered finite zero/pole endpoint pair supports at most one nondegenerate overlapping Plucker packet. Equivalently, this inversion is projective. For $E = [Z : W]$, write $c_j(E) = a_j W + b_j Z$. If $E_0, E_1 \in \mathbb{P}^1$ are the ordered double endpoints of H_0, H_1, then $P_0 \neq 0$ gives

$$\lambda_0 = \frac{c_1(E_0)}{c_0(E_0)}, \qquad \lambda_1 = 2\lambda_0 - \lambda_0^2 \frac{c_0(E_1)}{c_1(E_1)}.$$

Thus the projective endpoint pair, including infinity cases, determines the nondegenerate overlapping Plucker packet over a fixed initial row basis. If the two projective endpoints coincide, then the inversion formulas give $\lambda_1 = \lambda_0$. Consequently

$$c_2 - \lambda_1 c_1 = \lambda_0(c_1 - \lambda_0 c_0), \qquad H_1 = \lambda_0^2 H_0.$$

The corresponding norm H_1/H_0 is constant, so the diagonal projective endpoint case is charged to the constant-norm packet ledger rather than to the residual nonconstant square-norm branch. After the zero-row/proportional and constant-norm ledgers are removed, the nondegenerate overlapping chart therefore injects into ordered off-diagonal projective endpoint pairs. For an endpoint palette Ω, a fixed initial row basis supports at most $|\Omega|(|\Omega| - 1)$ nonconstant overlapping square-norm packets. In fact, if $A_i = \{E : c_i(E) = 0\}$, then $A_0 \neq A_1$, the first endpoint must avoid $A_0 \cup A_1$, and the second must avoid A_1 and the first endpoint. Conversely every such ordered pair recovers a unique nondegenerate packet. Hence the exact fixed-basis count is

$$|\Omega \setminus (A_0 \cup A_1)|\,(|\Omega \setminus A_1| - 1),$$

which is $(q-1)^2$ for $\Omega = \mathbb{P}^1(F_q)$. In the fixed-basis coordinate $x = c_1/c_0$, this is the elementary normal form $x_0 = \lambda_0 \in F^\times$,

$$x_1 = \frac{c_1(E_1)}{c_0(E_1)}, \qquad \lambda_1 = 2x_0 - \frac{x_0^2}{x_1},$$

with $x_1 \in F^\times$, $x_1 \neq x_0$, and

$$H_1 = -\frac{x_0^4}{x_1^2}(c_1 - x_1 c_0)^2.$$

The endpoint $x_1 = \infty$ gives $\lambda_1 = 2x_0$ and $H_1 = -x_0^4 c_0^2$. Equivalently the fixed-basis chart is exactly the slope-pair palette

$$\lambda_0 \in F^\times, \qquad \lambda_1 \in F \setminus \{\lambda_0\}.$$

If $\lambda_1 \neq 2\lambda_0$, then

$$x_1 = \frac{\lambda_0^2}{2\lambda_0 - \lambda_1},$$

while $\lambda_1 = 2\lambda_0$ is the endpoint $x_1 = \infty$. Thus the local nondegenerate branch has no hidden endpoint constraint after the initial row basis is fixed; the remaining M1 issue is global row-basis variation. In these slope-pair coordinates the square map itself is

$$m_{\lambda_0,\lambda_1} = \frac{(2\lambda_0 - \lambda_1)c_1 - \lambda_0^2 c_0}{c_1 - \lambda_0 c_0}, \qquad \frac{H_1}{H_0} = m_{\lambda_0,\lambda_1}^2.$$

It is a degree-one bijection from the open projective slope line to $F_q^\times$. Hence the full projective packet has exact multiplicative character balance, and passing to a finite-slope chart changes a nonprincipal character sum by at most the one omitted point at infinity. This fixed-basis branch has no local $\sqrt{q}$ Kummer loss. The same support view removes orientation duplicates. In the fixed-basis coordinate the square-packet palette depends only on the unordered support $\{x_0, x_1\} \subset \mathbb{P}^1(F_q)\backslash\{0\}$, since reversing the orientation replaces the degree-one parameter by a constant multiple of its inverse. Thus the distinct fixed-basis square-packet slope sets are bounded by

$$\frac{e}{2}\binom{q}{2} = \frac{e}{4}q(q-1),$$

for even quotient order e. The ordered endpoint-pair count remains a packet-incidence ledger, but the support-palette count is the sharper slope-set ledger. The slope-pair map to supports is exact: the support $\{\lambda_0, x_1\}$ has two preimages when both endpoints are finite nonzero, corresponding to the two orientations, and one preimage when the support contains ∞. Hence

$$2\binom{q-1}{2} + (q-1) = (q-1)^2$$

is precisely the ordered incidence count, while $\binom{q}{2}$ is the support count. No further local multiplicity is hidden in the slope-pair chart. The corresponding support-packet family is an exact incidence design on $\mathbb{P}^1(F_q)$: a point x lies in exactly $\binom{q}{2}$ packets if $x=0$, and exactly $\binom{q-1}{2}$ packets if $x \in \mathbb{P}^1(F_q)\setminus\{0\}$. Thus the full local packet family has no hidden point concentration; nonuniformity must come from the global row-basis/core-image selection of supports. More generally, for any selected support family Σ and any selected packet classes Θ_S, the packet incidence satisfies

$$I(x) \le \#\{S \in \Sigma : x \notin S\}, \qquad \sum_x I(x) = \frac{2(q-1)}{e}\sum_{S\in\Sigma}|\Theta_S|.$$

Equality in the first bound holds for the full class palette. Thus the global fixed-basis square-map problem reduces to the support-avoidance profile of the selected row-basis/core image. For full class palettes, if $m = |\Sigma|$ and $d_\Sigma(x)$ is the degree of x in the selected support graph on $U = \mathbb{P}^1(F_q)\setminus\{0\}$ with $d_\Sigma(0) = 0$, then

$$I_{\text{full}}(x) = m - d_\Sigma(x), \qquad \sum_x I_{\text{full}}(x)^2 = (q-3)m^2 + \sum_{x\in U} d_\Sigma(x)^2.$$

Thus full-palette concentration is exactly endpoint-degree energy of the selected support graph. Equivalently,

$$\sum_{x\in U} d_\Sigma(x)^2 = 2m + 2C_1(\Sigma),$$

where $C_1(\Sigma)$ is the number of unordered pairs of selected supports sharing exactly one endpoint. Thus the excess energy is precisely star-pair collision energy in the selected support image. If $\Delta = \max_x d_\Sigma(x)$, then

$$C_1(\Sigma) \le m(\Delta-1), \qquad \sum_x I_{\text{full}}(x)^2 \le (q-3)m^2 + 2m\Delta.$$

Consequently a full-palette second moment above $(q-3)m^2+2mD$ forces an endpoint star of degree $> D$. The same bound applies to arbitrary partial class selections, since their incidence function is pointwise bounded by the full-palette incidence:

$$\sum_x I(x)^2 \le (q-3)m^2 + 2m\Delta.$$

Thus partial square-coset choices do not create a new local source of second-moment mass. Writing $K = \sum_{S\in\Sigma}|\Theta_S|$, $M = 2(q-1)K/e$, and $B = |\{x : I(x) > 0\}|$, Cauchy's inequality gives

$$B \ge \frac{M^2}{(q-3)m^2 + 2m\Delta}.$$

More generally, if one tests an endpoint-degree cap D, then packing the same selected square-map packets into fewer than

$$\frac{M^2}{(q-3)m^2+2mD}$$

slope points forces some endpoint star of degree $> D$. Thus unexpectedly tight packing of many selected square-map packets forces the same endpoint-star alternative. This can be turned into a pruning reduction. Repeatedly remove all supports incident to an endpoint whose current degree is $> D$. The deleted supports are partitioned among endpoint-star templates, while the residual support family Σ_{ap} has maximum degree at most D. For the residual classes, with

$$K_{\mathrm{ap}} = \sum_{S\in\Sigma_{\mathrm{ap}}} |\Theta_S|, \quad M_{\mathrm{ap}} = 2(q-1)K_{\mathrm{ap}}/e, \quad B_{\mathrm{ap}} = |\{x : I_{\mathrm{ap}}(x) > 0\}|,$$

one has

$$B_{\mathrm{ap}} \geq \frac{M_{\mathrm{ap}}^2}{(q-3)m_{\mathrm{ap}}^2 + 2m_{\mathrm{ap}}D} \qquad (m_{\mathrm{ap}} = |\Sigma_{\mathrm{ap}}| > 0).$$

Thus after endpoint stars are charged to quotient/low-template ledgers, the aperiodic residual cannot be overpacked by local square-map geometry. Equivalently, for any residual slope budget R, either $B_{\mathrm{ap}} > R$ or

$$\left(\frac{2(q-1)}{e}\right)^2 K_{\mathrm{ap}}^2 \leq R\big((q-3)m_{\mathrm{ap}}^2 + 2m_{\mathrm{ap}}D\big),$$

or, in average-density form,

$$(K_{\mathrm{ap}}/m_{\mathrm{ap}})^2 \leq \frac{R\big((q-3) + 2D/m_{\mathrm{ap}}\big)}{(2(q-1)/e)^2}.$$

Thus a dense bounded-degree residual must occupy many slopes; if it does not, the residual square-coset class selection is sparse rather than geometrically overpacked. In the full-palette case there is an exact sharper support formula:

$$\{x : I_{\mathrm{full}}(x) > 0\} = \mathbb{P}^1(F_q) \setminus \bigcap_{S\in\Sigma} S, \qquad B_{\mathrm{full}} = q+1-\left|\bigcap_{S\in\Sigma} S\right|.$$

Hence, after support-star pruning, if $m_{\mathrm{ap}} > D$, the residual common intersection is empty and the full-palette residual covers all $q+1$ projective slopes. Any small pruned residual support therefore comes from at most D remaining supports or from genuinely partial square-coset palettes. The partial-palette leftover has an exact defect ledger. If Θ_S^c denotes the missing square-coset classes and

$$J(x) = \#\{(S,\theta) : \theta \in \Theta_S^c,\ x \in P_S(\theta)\},$$

then

$$I(x) + J(x) = m - d_\Sigma(x), \qquad \sum_x J(x) = \frac{2(q-1)}{e} \sum_{S\in\Sigma} |\Theta_S^c|.$$

Thus, after pruning with maximum residual degree D and $m_{\mathrm{ap}} > D$, the number of uncovered residual slopes satisfies

$$|\{x : I_{\mathrm{ap}}(x) = 0\}| \leq \frac{2(q-1)}{e} \frac{\sum_{S\in\Sigma_{\mathrm{ap}}} |\Theta_S^c|}{m_{\mathrm{ap}} - D}.$$

So a large bounded-degree residual can leave many slopes uncovered only by spending many missing local square-coset classes. Combining the two residual ledgers gives the local support-budget alternative. For any $0 \le R \le q+1$, after pruning endpoint stars with degree cap D, either $m_{\mathrm{ap}} \le D$, or $B_{\mathrm{ap}} > R$, or both

$$\left(\frac{2(q-1)}{e}\right)^2 K_{\mathrm{ap}}^2 \le R((q-3)m_{\mathrm{ap}}^2 + 2m_{\mathrm{ap}}D)$$

and

$$(m_{\mathrm{ap}} - D)(q+1-R) \le \frac{2(q-1)}{e} C_{\mathrm{ap}}$$

hold. Thus small residual slope support must be paid for by both selected class sparsity and missing-class mass. Setting $R = B_{\mathrm{ap}}$ gives a direct palette-density lower bound. With $h = e/2$,

$$\alpha_{\mathrm{ap}} = K_{\mathrm{ap}}/(hm_{\mathrm{ap}}), \qquad \gamma_{\mathrm{ap}} = C_{\mathrm{ap}}/(hm_{\mathrm{ap}}) = 1 - \alpha_{\mathrm{ap}},$$

and $m_{\mathrm{ap}} > D$,

$$B_{\mathrm{ap}} \ge \frac{(q-1)^2\alpha_{\mathrm{ap}}^2 m_{\mathrm{ap}}^2}{(q-3)m_{\mathrm{ap}}^2 + 2m_{\mathrm{ap}}D}, \qquad B_{\mathrm{ap}} \ge q+1 - \frac{(q-1)\gamma_{\mathrm{ap}} m_{\mathrm{ap}}}{m_{\mathrm{ap}} - D}.$$

In particular, if $m_{\mathrm{ap}} \ge 2D$ and $\gamma_{\mathrm{ap}} \le \varepsilon$, then $B_{\mathrm{ap}} \ge q+1-2\varepsilon(q-1)$. Equivalently, for a target support budget R, if $m_{\mathrm{ap}} > D$ and either

$$\left(\frac{2(q-1)}{e}\right)^2 K_{\mathrm{ap}}^2 > R((q-3)m_{\mathrm{ap}}^2 + 2m_{\mathrm{ap}}D)$$

or

$$\frac{2(q-1)}{e} C_{\mathrm{ap}} < (m_{\mathrm{ap}} - D)(q+1-R),$$

then $B_{\mathrm{ap}} > R$. Thus a global proof can close this local branch by showing enough selected palette density or too few missing palette classes. The local fixed-basis square-map output is therefore a four-way theorem: after greedy endpoint-star pruning at cap D, the deleted part is charged to endpoint-star templates; or $m_{\mathrm{ap}} \le D$; or $B_{\mathrm{ap}} > R$; or the small-support residual satisfies both support-budget inequalities above. The last case is excluded by either displayed strict density trigger, so the only remaining small-support branch is a genuine global palette-sparsity certificate. Equivalently, the optional certificate is the exact finite object

$$m_{\mathrm{ap}} > D, \quad B_{\mathrm{ap}} \le R, \quad \max_x d_{\Sigma_{\mathrm{ap}}}(x) \le D,$$

together with the two support-budget inequalities. Thus the next global M1 task is to rule out this sparse certificate from the row-basis/core image, or to identify it as quotient/template structure. With $h = e/2$, $\alpha = K/(hm)$, and $\gamma = C/(hm) = 1 - \alpha$, any such certificate lies in the feasibility window

$$\alpha^2 \le \frac{R((q-3)m^2 + 2mD)}{(q-1)^2 m^2}, \qquad \gamma \ge \frac{(m-D)(q+1-R)}{(q-1)m}.$$

So a global lower bound on selected palette density above this window rules out the sparse certificate in the corresponding parameter range. If additionally $m \ge LD$ for an integer $L \ge 2$, this becomes the uniform far-from-star window

$$\alpha^2 \le \frac{R(q-3+2/L)}{(q-1)^2}, \qquad \gamma \ge \frac{L-1}{L}\,\frac{q+1-R}{q-1}.$$

Thus certificates whose residual size is a fixed factor beyond the endpoint degree cap can be excluded by a single L-dependent density threshold. Contrapositively, if either strict density gap violates this window, then

$$m < LD.$$

The residual then has fewer than LD two-point supports and endpoint footprint $< 2LD$. Hence a global row-basis/core-image density theorem above the displayed far-from-star window reduces every remaining small-support residual to a bounded near-star/template ledger. Writing $h = e/2$ and

$$s_{LD} = \min(q+1, \max(0, 2LD-1)),$$

the number of such residual palette templates is bounded by

$$\sum_{s=0}^{s_{LD}} \binom{q+1}{s} 2^{h\binom{s}{2}} \leq (s_{LD}+1)(q+1)^{s_{LD}} 2^{h\binom{s_{LD}}{2}}.$$

For fixed D, L, e, the near-star branch is therefore a polynomial-size endpoint-template ledger. Equivalently, define

$$\theta_L(q,R) = \min\left(\frac{\sqrt{R(q-3+2/L)}}{q-1}, 1 - \frac{L-1}{L}\frac{q+1-R}{q-1}\right).$$

If a global row-basis/core-image argument proves $\alpha_{\mathrm{ap}} > \theta_L(q,R)$ for every sparse residual output, then the far-from-star sparse branch is empty. The local output reduces to endpoint stars, D-small residuals, large support $B_{\mathrm{ap}} > R$, and the polynomial-size near-star template ledger above. Moreover, if $L_E(z) = Wz - Z$ for $E = [Z : W]$, then in the off-diagonal case

$$\frac{H_1}{H_0} = \gamma^2 \left(\frac{L_{E_1}}{L_{E_0}}\right)^2$$

for some $\gamma \in F^\times$. Thus this residual chart is exactly a degree-one square-map packet with divisor $2E_1 - 2E_0$, up to an overall square scalar. For even character order e, fixing the ordered pair E_0, E_1 leaves only the $e/2$-packet square-coset palette

$$P_{E_0,E_1}(\theta) = \{z : \chi(L_{E_1}(z)/L_{E_0}(z))^2 = \theta\}, \qquad \theta \in 2\mathbb{Z}/e\mathbb{Z},$$

which partitions the open slope line away from E_0, E_1. Combining the endpoint-pair injection and this palette, a fixed initial row basis and endpoint palette Ω contribute at most $(e/2)|\Omega|(|\Omega|-1)$ distinct nonconstant slope sets; for the full projective line over F_q, this is $(e/2)q(q+1)$. Each such packet has at most $2(q-1)/e$ finite slopes. If both endpoints are finite, exactly one packet loses the point at infinity and has size $2(q-1)/e-1$; otherwise all packets have the full $2(q-1)/e$ finite slopes. The finite endpoints are also ledger charges rather than residual live slopes. If $B_R(z_0) = 0$ and $Q_R(z_0) \neq 0$, then $p_R(z_0) = B_R(z_0)/Q_R(z_0) = 0$, so the recovered locator lies on the fixed-root line $p = 0$. If $Q_R(z_0) = 0$, then z_0 is outside the recurrence chart; whenever the recurrence is solvable, the denominator-zero classification gives the existing full-plane or fixed-root charge, and if it is not solvable there is no formal two-root extension. The endpoint at infinity is the same statement in the reciprocal slope chart. Consequently, for any finite family $\mathcal{S}$ of degree-one square-map packets, the number of distinct slope sets in $\mathcal{S}$ is at most

$$\frac{e}{2}\#\{\Pi(M) : P(M,\kappa) \in \mathcal{S}\}.$$

In particular a single nonconstant square-norm branch contributes at most $e/2$ distinct square-map slope sets; repeated square roots and parallel/inverse parameters are multiplicity inside the same endpoint palette, not a new slope-growth source. This is cruder than the cover bound below, but it is genus-free; the cover sums are only needed for the extra factor-e saving. Now define

$$S_a^+ = \sum_{C_R^\times} \chi^a(r_+), \qquad S_{a,b} = \sum_{C_R^\times} \chi^a(r_+)\chi^b(r_-).$$

The ordered mixed-domain count has the exact expansion

$$N_R^{+,-}(D) = \frac{1}{e}\sum_a S_a^+ - \frac{1}{e^2}\sum_{a,b} S_{a,b} + O(1).$$

Its principal contribution is $((e-1)/e^2)|C_R^\times| + O(1)$. If no nonprincipal term is geometrically trivial, the standard curve Kummer-Weil input gives

$$N_R^{+,-}(D) \le \frac{e-1}{e^2}|C_R^\times| + C_e\sqrt{p} + O_e(1).$$

The norm map gives a necessary condition for any cover-level power branch: if $F_{a,b} = r_+^a r_-^b$ has divisor divisible by e on the cover, then pushing forward by $C_R \to \mathbb{P}_z^1$ gives

$$\operatorname{div}\big((B_R/Q_R)^{a+b}\big) \equiv 0 \pmod{e}.$$

Hence, after the constant-norm and square-norm branches above are charged or isolated, all one-root terms and all pair terms with $a+b \not\equiv 0 \pmod{e}$ are nontrivial bounded-conductor cover sums. The only remaining cover-level power target is the anti-diagonal ratio term $\chi^a(r_+/r_-)$. This target is almost closed as well. Let $\Phi_R = r_+/r_-$. Since $(A_R + Y)(A_R - Y) = 4Q_R B_R$, the map $\Phi_R : C_R \to \mathbb{P}^1$ has degree at most four, and

$$\Phi_R/N_R = r_-^{-2}, \qquad N_R = B_R/Q_R.$$

Thus even-order anti-ratio power branches are already square-norm branches, while odd orders $m \ge 5$ are impossible for nonconstant Φ_R. After the square-norm branch is isolated, the only possible nonconstant anti-ratio power branch is the cubic one $\operatorname{div}_C(r_+/r_-) \equiv 0 \pmod{3}$. In particular, if $3 \nmid e$, no cover-level power branch remains after the norm exceptions. If $3 \mid e$, geometric triviality of this cubic term would require $\Phi_R = cG^3$ in $k(C_R)^*$. Since $\deg \Phi_R \le 4$, nonconstant Φ_R then has $\deg \Phi_R = 3$ and $\deg G = 1$. Thus a genuine cubic no-cancellation term forces the normalization of C_R to be rational; on a geometrically integral genus-one cover the cubic anti-ratio term is still geometrically nontrivial and contributes only a bounded-conductor Weil term. On the rational cubic branch, the only possible large anti-diagonal coefficients are $a = e/3, 2e/3$, when $3 \mid e$. Hence, after the constant-norm and square-norm exceptions are removed, their conjugate cubic constants contribute at worst $+|C_R^\times|/e^2$ in the mixed-domain expansion, and

$$N_R^{+,-}(D) \le \frac{1}{e}|C_R^\times| + C_e\sqrt{p} + O_e(1) \le |D| + C_e\sqrt{p} + O_e(1)$$

for geometrically integral rational normalization. The diagonal pair terms in this expansion descend to the slope line:

$$S_{a,a} = \sum_z (1 + \chi_2(\Theta_R(z)))\chi^a(B_R(z)/Q_R(z)),$$

on the open locus $Q_R B_R \neq 0$. Hence each diagonal term is a sum of two bounded-support $\mathbb{P}^1$ Kummer traces, with support contained in the roots of B_R, Q_R, Θ_R and infinity. Its only no-cancellation branches are the explicit power-divisor congruences for $(B_R/Q_R)^a$ and $(B_R/Q_R)^{a\ell/e}\Theta_R^{\ell/2}$, $\ell = \operatorname{lcm}(e,2)$. Only the off-diagonal $S_{a,b}$, $a \neq b$, and the one-root sums S_a^+ remain genuinely cover-level terms. For index two, these cover-level terms cancel. Writing $\rho = \chi$, sheet symmetry gives $S_1^+ = S_1^-$, hence

$$N_R^{+,-}(D) = \frac{1}{4}\left(|C_R^\times| - S_{1,1}\right) + O(1) = \frac{1}{4}|C_R^\times| - \frac{1}{4}\sum_z (1+\rho(\Theta_R(z)))\rho(B_R(z)/Q_R(z)) + O(1).$$

Thus quadratic-residue domains reduce the mixed root-core count entirely to bounded-support $\mathbb{P}^1$ Kummer sums after the fixed-zero-root branch is charged. If neither B_R/Q_R nor $\Theta_R B_R/Q_R$ is a square divisor, the nonsplit branch is at most $|D|/2 + C\sqrt{p} + O(1)$, and the split-square branch is at most $|D| + C\sqrt{p} + O(1)$. Thus a geometrically integral nonsplit quartic-cover branch has at most $(1-1/e)|D| + C_e\sqrt{p} + O_e(1)$ points, with conjugate or no-point degeneracies only smaller, while the split-square rational branch has at most $2(1-1/e)|D| + C_e\sqrt{p} + O_e(1)$. Each unordered mixed split pair contributes to exactly one ordered sheet, and repeated roots contribute nothing to the mixed-domain condition. Combining the cover-power reductions gives the classified per-core form: after fixed-zero-root and constant-norm branches are charged and the negative-parity rational one-root-square subbranch is isolated, positive-parity square norm satisfies $(e-2)|D| + C_e\sqrt{p} + O_e(1)$, geometrically integral genus-one branches satisfy $(1-1/e)|D| + C_e\sqrt{p} + O_e(1)$, negative-parity square-norm genus-one branches satisfy $|D| + C_e\sqrt{p} + O_e(1)$, geometrically integral rational cubic branches satisfy $|D| + C_e\sqrt{p} + O_e(1)$, rational one-root-square branches satisfy $O_e(\sqrt{p})$ for the negative root-square sign and $2|D| + C_e\sqrt{p} + O_e(1)$ for the positive root-square sign, and split-square rational branches satisfy $2(1-1/e)|D| + C_e\sqrt{p} + O_e(1)$. The index-two nonsplit case away from the two square gates has the sharper $|D|/2 + C\sqrt{p} + O(1)$ bound above.

Consequently, after fixed-slope root slices are removed, the residual $t = 2$ one-exchange graph has maximum degree at most j: each locator has only j possible $(j-1)$-cores, and each core supplies at most one residual neighbor. In particular the unordered residual edge count is at most $j|A|/2$, and also at most $\binom{|D|}{j-1}$ by counting cores. Plugging this into the average support-collinearity ledger `experimental/notes/m1/m1_average_support_collinearity.md`, and writing the line-field size as Q, gives for the residual family A_{res}, $M = |A_{\text{res}}|$,

$$B_2^{\max}(A_{\text{res}}) \leq \frac{1-p_z}{Mp_z} + \frac{4jQ}{M}, \qquad p_z = Q^{-2}(1-Q^{-2}).$$

Thus this branch contributes $o(1)$ to the average missing-slope density whenever $Mp_z \to \infty$ and $M/(jQ) \to \infty$. The remaining M1 obstruction must come from small residual family size, root-slice charges, or higher packet/two-exchange structure; this is not yet a worst-case packing theorem.

The same substitution has a packet-level higher-exchange form. Inside one same-slope affine h-exchange packet, after full moving r-root fibers have been charged, the residual codegree bound $\Gamma_r \leq \binom{h}{r}(Q_F^{r-1} - 1)$ gives, for average-ledger support slack τ,

$$B_\tau^{\max}(A_{\text{res}}) \leq \frac{1-p_z}{Mp_z} + \frac{4}{M}\sum_{r=1}^{\min(h,\tau-1)} \binom{h}{r}(Q_F^{r-1}-1)Q_F^{\tau-r}, \qquad p_z = Q_F^{-\tau}(1-Q_F^{-\tau}).$$

This is still a local packet ledger input; a global M1 estimate must also sum over packet choices and the quotient, tangent/contained, split-root, and fixed-slope ledgers.

19.5 M1: variable-line packet reduction

The follow-up variable-line packet note isolates a local reduction inside the same Hankel-pencil branch. Fix a $(j-2)$-core R and a non-fixed proper one-dimensional determinant component L in the two-root plane

$$T = R \cup \{x, y\}, \qquad s = x + y, \quad p = xy.$$

The constant-line collapse above verifies the needed variable-slope hypothesis for every surviving fixed-sum or nondegenerate product-Mobius packet after fixed-root and full-plane charges. Thus the active-new mass r_L on such an L obeys

$$r_L \le \mathbf{1}_{d_L=m_L=r_L=1} + (d_L - m_L) + 2\binom{m_L}{2}.$$

The unordered packet pairs in the final term inject into the global different-slope two-exchange codegree ledger, so summing over non-fixed line packets gives

$$\sum_L r_L \le S_{\mathrm{dom}} + Q_{\mathrm{def}} + 2E_{\mathrm{pkt}}.$$

The remaining domain-singleton term is reduced to contained/tangent and one-outside target images:

$$S_{\mathrm{dom}} \le Z_0 + 2\binom{j}{2}\binom{n-j}{2}|\mathrm{Target}_{\mathrm{cb}}| + (j-1)\binom{n-j+1}{2}|\mathrm{Target}_{\mathrm{off}}|.$$

Writing $a = n - j$, the exceptional zero-lower term vanishes whenever

$$a > \frac{n+1}{2},$$

and hence throughout the positive-slack rate-half range $k \ge n/2$, $a = k + t$, $t \ge 1$. Consequently the non-fixed variable-line branch is reduced to two explicit residual estimates: the active different-slope two-exchange codegree and the one-outside boundary shadow image, after quotient defects, contained/tangent ledgers, and fixed-slope boundary slices have been paid. Indeed the shadow-fiber bound gives

$$|\mathrm{Target}_{\mathrm{off}}^{\mathrm{res}}| \le 2|\mathrm{Shadow}_{\mathrm{off}}^{\mathrm{res}}|.$$

After the fixed-anchor boundary-core slices are also charged, the sharper fiber bound

$$(j-1)|\mathrm{Target}_{\mathrm{off}}^{\mathrm{res}}| \le 2|\mathrm{Core}_{\mathrm{off}}^{\mathrm{res}}|$$

may be substituted directly into the singleton target term. The boundary contribution

$$(j-1)\binom{n-j+1}{2}|\mathrm{Target}_{\mathrm{off}}^{\mathrm{res}}|$$

is then bounded by

$$2\binom{n-j+1}{2}|\mathrm{Core}_{\mathrm{off}}^{\mathrm{res}}|,$$

so a polynomial bound for the boundary-core image closes this branch with only an n^2 bookkeeping loss, rather than the older n^3 boundary-image loss. The fixed-core graph form also gives

$$|\mathrm{Core}_{\mathrm{off}}^{\mathrm{res}}| \le 2(n-j+2)|\mathrm{Root}_{\mathrm{off}}^{\mathrm{res}}|,$$

so bounding the active $(j-2)$-core image closes the same branch with an n^3 boundary loss after all boundary-core root slices have been charged.

Status. This is a proved local packet lemma under the Hankel-pencil normal form after the same-slope, fixed-root, full-plane, and boundary-slice charges recorded above. It is not the final all-line M1 theorem and does not add a leaderboard row.

19.6 M1: width-one fixed-root closure

PR #136, due to AllenGrahamHart, extracts a compact closure criterion for the width-one critical-tail branch. At a base-free canonical $b = 2$ node, let

$$V = \langle P, Q \rangle \subset F[X]_{<q}, \qquad |D^{can}| = q + s.$$

A root-free width-one certificate is a direction $A \in V$ whose root shadow on D^{can} has maximal size $q-1$. Writing $Z = Z(A)$ and $O = D^{can} \setminus Z$, one has

$$A = c\ell_Z, \qquad |Z| = q-1, \qquad |O| = s+1.$$

Equivalently, O supports a width-one certificate if and only if

$$L_O = \ell_{D^{can} \setminus O} \in V,$$

or equivalently all 3×3 coefficient minors of P, Q, L_O vanish. Thus width-one certificates are bounded-complement rank tests rather than an unstructured projective-direction family.

The large width-one cube descends losslessly under fixed-root absorption. If $P_m \subset Z$, then

$$A^{P_m} = A/\ell_{P_m} = c\ell_{Z \setminus P_m}, \qquad D^{can} \setminus P_m = (Z \setminus P_m) \sqcup O,$$

so the descended width remains one and the bounded complement O is unchanged. With $a = \lfloor (q-2)/2 \rfloor$, the co-small flag cube has the first-root partition

$$\sum_{e=1}^{a} \binom{q-1}{e} = \sum_{x \in Z} \sum_{f=0}^{a-1} \binom{|Z_{>x}|}{f}.$$

After exposing the first root x, every nonempty width-one flag injects into a one-root absorbed fixed-divisor/root-slice family. Quantitatively, for

$$M_x = \#\{F \subset Z_{>x} : |F| \le a-1\},$$

one gets

$$\max_{x \in Z} M_x \ge \frac{1}{q-1} \sum_{e=1}^{a} \binom{q-1}{e} \ge \frac{2^{q-1}}{2q^2} \qquad (q \ge 4).$$

Thus a large width-one cube already produces a large one-root slice; it cannot hide as a many-first-root averaging phenomenon.

Consequently, in fixed surplus $s_0 \le \sigma$, small nodes are polynomial by the active-tree bound, and on $q_S > s_S + 2$ there is at most one width-one shadow per node. The local closure criterion is

$$WO_1(A_0) \le FixedRootOneRoot_{r1}(A_0) + O_\sigma(Q^{\sigma+1}),$$

where Q is the initial quotient width. This closes the width-one critical-tail branch only after a polynomial bound for the displayed one-root fixed-root/root-slice ledger is proved or imported. It is not the final all-line M1 theorem and does not add a leaderboard row.

19.7 Challenge-map pullbacks and public status corrections

The adjacent-ledger update adds the missing pullback rule from slope ledgers to protocol challenge maps. For a challenge-to-slope map $\phi : K \to Z$, a bad set $B \subseteq Z$ pulls back with probability

$$\frac{|\phi^{-1}(B)|}{|K|}.$$

If the nonempty fibers have sizes $s_1 \geq s_2 \geq \cdots$, the adversarial envelope for $|B| = L$ is $(s_1 + \cdots + s_L)/|K|$. For an $\mathbb{F}_q$-linear map of rank r, the effective denominator is q^r, not the ambient challenge space size. This prevents accidental multiplication of denominators in protocol ledgers.

The public-board status corrections are:

row	a	tangent floor	status
Cycle116 gate	262	251	gate unconditional; exact count conditional
Cycle119 gate	263	250	gate unconditional; exact count conditional
reserve272	272	241	proved but subsumed
reserve288	288	225	proved but subsumed
reserve313	313	200	proved but subsumed

The exact Cycle84 numerator 52,747,567,092 remains a computation-dependent record; only the weaker $> 2^{-128}$ gate follows from the generic tangent floor without that finite census.

20 High-Agreement Tangent Staircase

Status: New-local finite MDS theorem. This section records a generic moving-root tangent floor and, in the very-high-agreement range, a matching upper bound. It is independent of quotient-periodic structure and does not use smoothness. It is relevant to the finite row ledger because the bad slopes are paid with the actual line field q_{line}.

For a polynomial $P \in F[X]$, write

$$P = P_{<k} + P_{\geq k}, \qquad \deg P_{<k} < k,$$

where $P_{\geq k}$ is the sum of the terms of degree at least k. Let $\mathrm{LD}_{\text{sw}}(C, a)$ denote the largest possible number of finite slopes on a received line that have a support-wise noncontained explanation on some support of size at least a.

Theorem 6 (moving-root tangent floor)**.** *Let $D \subset F$ have $|D| = n$, and let $C = \mathrm{RS}[F, D, k]$. For every integer a with $k + 1 \leq a \leq n$,*

$$\mathrm{LD}_{\text{sw}}(C, a) \geq n - a + 1.$$

Proof. Choose a subset $A \subset D$ of size $a - 1$, and put

$$B_A(X) = \prod_{x \in A} (X - x).$$

Define the received line

$$f = (XB_A)_{\geq k}, \qquad g = -(B_A)_{\geq k}.$$

For every $t \in D \setminus A$, set

$$L_t(X) = (X - t)B_A(X).$$

Then L_t is the locator of the support $A \cup \{t\}$, which has size a, and linearity of truncation gives

$$f + tg = (L_t)_{\geq k}.$$

On every root of L_t, the word represented by $(L_t)_{\geq k}$ agrees with $-(L_t)_{<k}$, a codeword of C. Thus the slope t is explained on $A \cup \{t\}$.

It remains to check support-wise noncontainment on this same support. On A, since $B_A = 0$, we have

$$g = -(B_A)_{\geq k} = (B_A)_{<k}.$$

If g were explained on $A \cup \{t\}$ by a degree-$< k$ polynomial q, then $q = (B_A)_{<k}$, because $|A| = a - 1 \geq k$. At the remaining point t, this would force

$$-(B_A)_{\geq k}(t) = (B_A)_{<k}(t),$$

that is, $B_A(t) = 0$, contrary to $t \notin A$. Hence g is not separately explained on $A \cup \{t\}$, so the slope is support-wise noncontained. The slopes $t \in D \setminus A$ are distinct, giving $n - a + 1$ bad finite slopes. □

Theorem 7 (very-high-agreement tangent staircase)**.** *Let* $C = \mathrm{RS}[F, D, k]$ *with* $|D| = n$*. If*

$$3a - 2n \geq k,$$

then

$$\mathrm{LD}_{\mathrm{sw}}(C, a) = n - a + 1.$$

Proof. The lower bound is Theorem 6. For the upper bound, put $s = n - a$. If a received line has at most one support-wise noncontained finite slope at agreement at least a, then the claimed upper bound is immediate. Otherwise pick two such slopes $z_1 \neq z_2$, with explaining codewords $p_1, p_2 \in C$ on supports S_1, S_2, respectively. Since $|S_i| \geq a$,

$$|S_1 \cap S_2| \geq 2a - n.$$

On this intersection,

$$g = (p_1 - p_2)/(z_1 - z_2), \qquad f = p_1 - z_1 g.$$

The condition $3a - 2n \geq k$ and the inequality $a \leq n$ imply $2a - n \geq k$. Therefore these two right-hand sides extend uniquely to codewords $c_g, c_f \in C$. Let S_0 be a maximal common support on which $f = c_f$ and $g = c_g$, and put $b = |S_0|$ and $d = n - b$. Then

$$b \geq 2a - n, \qquad d \leq 2(n - a) = 2s.$$

By maximality, outside S_0 there is no coordinate at which both residuals $f - c_f$ and $g - c_g$ vanish.

The residual-budget proposition, Proposition 7, is applicable because

$$a + b - n \geq a + (2a - n) - n = 3a - 2n \geq k.$$

Here its common-zero parameter is $c_0 = 0$. If $b \geq a$, then every noncontained slope needs at least one residual zero outside S_0, and there are only $d \leq s$ such coordinates. If $b < a$, then every noncontained slope needs at least

$$h = a - b = d - s$$

private residual zeros outside S_0. Since $s < d \leq 2s$, the residual budget gives

$$\#\{\text{bad slopes}\} \leq \left\lfloor \frac{d}{d - s} \right\rfloor \leq s + 1 = n - a + 1.$$

This proves the matching upper bound. □

Corollary 2 (tangent-star extremizers in the exact range)**.** *Let $C = \mathrm{RS}[F, D, k]$, $|D| = n$, and suppose that*

$$k + 1 \leq a \leq n, \qquad 3a - 2n \geq k.$$

If a received line $f + zg$ has exactly $n - a + 1$ support-wise noncontained finite bad slopes at agreement at least a, and if $n - a + 1 \geq 2$, then the line is tangent-star extremal in the following sense. There are codewords $c_f, c_g \in C$ and a common support $S_0 \subseteq D$ of size $a - 1$ such that, after replacing f, g by $f - c_f, g - c_g$, every bad slope is obtained by choosing exactly one residual coordinate in $D \setminus S_0$. More precisely, writing $\Omega = D \setminus S_0$, the map

$$x \longmapsto -\frac{f(x) - c_f(x)}{g(x) - c_g(x)}$$

is a bijection from Ω to the bad-slope set.

Proof. Choose two distinct bad slopes and form the maximal common code-line support S_0 as in the proof of Theorem 7; write $b = |S_0|$, $s = n - a$, and $d = n - b$. Equality in the upper bound from that proof cannot occur when $b \geq a$, because then there are at most $n - a$ bad slopes. Hence $b < a$. The residual-budget bound is

$$\#\{\text{bad slopes}\} \leq \left\lfloor \frac{d}{d - s} \right\rfloor \leq s + 1 = n - a + 1.$$

For equality, the second inequality must be sharp, so $d - s = 1$, equivalently $b = a - 1$. The maximality of S_0 gives no common residual zeros outside S_0. Since $|\Omega| = n - a + 1$ and each residual coordinate can vanish for at most one finite slope, equality forces every residual coordinate to be used exactly once. This is precisely the displayed residual-root bijection. ☐

Corollary 3 (exact high-agreement gate for the $\mathbb{F}_{17^{32}}$, rate-1/2 row)**.** *Let $H \subset \mathbb{F}_{17^{32}}^*$ be any multiplicative subgroup of order* 512*, and let*

$$C = \mathrm{RS}[\mathbb{F}_{17^{32}}, H, 256].$$

For every $a \geq 427$,

$$\mathrm{LD}_{\mathrm{sw}}(C, a) = 513 - a.$$

In particular,

$$\mathrm{LD}_{\mathrm{sw}}(C, 506) = 7, \qquad \mathrm{LD}_{\mathrm{sw}}(C, 507) = 6.$$

Since

$$6 \cdot 2^{128} < 17^{32} < 7 \cdot 2^{128},$$

we have

$$\frac{\mathrm{LD}_{\mathrm{sw}}(C, 507)}{17^{32}} < 2^{-128} \quad \textit{but} \quad \frac{\mathrm{LD}_{\mathrm{sw}}(C, 506)}{17^{32}} > 2^{-128}.$$

Thus, under the finite-slope support-wise MCA convention, the 2^{-128} integer agreement staircase for this row is pinned between 506 *and* 507*. The largest safe integer Hamming radius is* 5*, while integer radius* 6 *is already unsafe. The corresponding normalized grid radii are*

$$5/512 \quad \textit{and} \quad 6/512 = 3/256.$$

As a real-radius supremum, the transition is $3/256$, but the closed endpoint $\delta = 3/256$ is on the unsafe side.

Proof. Here $n = 512$ and $k = 256$. The hypothesis $3a - 2n \geq k$ becomes $3a - 1024 \geq 256$, hence $a \geq 427$. The exact formula is Theorem 7. The arithmetic inequalities are

$$17^{32} - 6 \cdot 2^{128} = 326217393234836465063858652730341978625 > 0$$

and

$$7 \cdot 2^{128} - 17^{32} = 14064973686101998399515954701426232831 > 0.$$

$\square$

Corollary 4 (no seventh finite-slope branch past agreement 506)**.** *For $C = \mathrm{RS}[\mathbb{F}_{17^{32}}, H, 256]$ with $|H| = 512$, no finite-slope support-wise MCA mechanism can add a seventh bad slope at agreement $a \geq 507$. At equality, any extremal line is tangent-star in the sense of Corollary 2.*

Proof. By Corollary 3,

$$\mathrm{LD}_{\mathrm{sw}}(C, a) = 513 - a \qquad (a \geq 427).$$

For every $a \geq 507$, the right-hand side is at most 6. Thus a seventh finite support-wise bad slope is impossible in this range. The extremizer classification is Corollary 2. $\square$

Remark 6 (effect on the earlier agreement-353 target)**.** *The agreement-353 target is no longer a frontier for the rate-*$1/2$*,* $\mathbb{F}_{17^{32}}$*,* $n = 512$ *row: Theorem 6 already gives*

$$\mathrm{LD}_{\mathrm{sw}}(C, 353) \geq 512 - 353 + 1 = 160.$$

The quotient-core strict352 packet remains a useful mechanism record, but the generic tangent floor gives a stronger lower bound at agreement 352 *and beyond. Corollary 4 rules out a seventh finite-slope branch past agreement* 506*. The adjacent-ledger section below treats the high-agreement projective-slope, CA, curve-MCA, and interleaved-list coding objects; protocol-facing conversions still require separate challenge-field, extension-lift, folding, query, and cryptographic ledgers.*

21 High-Agreement Adjacent Ledgers

Status: New-local finite Reed–Solomon theorem package, one MDS interleaved-list theorem, and a Conditional protocol-ledger corollary. The tangent staircase of Section 20 settles more than finite-slope support-wise MCA. In the same high-agreement range it also pins the no-loss correlated-agreement line count, the projective-slope line count, the finite-parameter degree-d curve count, and the interleaved-list size. The protocol statement at the end of this section is only a ledger corollary: it applies to reductions whose coding error is exactly one of the listed terms, plus an explicitly added query/folding error.

Except in the interleaved-list theorem, $C = \mathrm{RS}[F, D, k]$ with $|D| = n$, dimension k, and $q = |F|$. The interleaved-list uniqueness statement only uses the MDS property. The agreement parameter is an integer a, so the corresponding closed Hamming radius is $1 - a/n$.

21.1 No-loss CA and projective slopes

Let $\mathrm{LD}_{\mathrm{ca}}(C, a)$ denote the maximum number of finite slopes $z \in F$ on an affine received line $f + zg$ such that $f + zg$ has a C-explanation on at least a positions while the pair (f, g) has no C^2-explanation on any support of size at least a. This is the integer-agreement version of no-loss CA. Let $\mathrm{LD}_{\mathrm{sw,proj}}(C, a)$ be the corresponding support-wise MCA count when slopes are points of $\mathbb{P}^1(F)$, so the point at infinity is also allowed.

Theorem 8 (high-agreement CA and projective tangent staircase)**.** *Let* $C = \mathrm{RS}[F, D, k]$ *with* $|D| = n$*. If*

$$3a - 2n \geq k,$$

then

$$\mathrm{LD}_{\mathrm{ca}}(C, a) = \mathrm{LD}_{\mathrm{sw,proj}}(C, a) = n - a + 1.$$

Proof. The finite-slope support-wise count is $n - a + 1$ by Theorem 7. Every no-loss CA-bad finite slope is support-wise MCA-bad: if the line word is explained on a support S of size at least a and (f, g) were simultaneously explained on that same S, then (f, g) would be C^2-close at the no-loss CA radius. Thus

$$\mathrm{LD}_{\mathrm{ca}}(C, a) \leq n - a + 1.$$

The moving-root line from Theorem 6 gives the matching CA lower bound. Indeed, with $A \subset D$, $|A| = a - 1$, put $B_A(X) = \prod_{x \in A}(X - x)$,

$$f = (XB_A)_{\geq k}, \qquad g = -(B_A)_{\geq k}.$$

For each $t \in D \setminus A$, the slope t is explained on $A \cup \{t\}$. Suppose (f, g) were explained by two degree-$< k$ polynomials on some support T of size at least a. Since

$$|T \cap A| \geq a - (n - a + 1) = 2a - n - 1,$$

and the hypothesis $3a - 2n \geq k$ implies $2a - n - 1 \geq k$ when $a < n$ (the case $a = n$ gives $n - 1 \geq k$), the two explaining polynomials must be the unique degree-$< k$ polynomials agreeing with f and g on A, namely $-(XB_A)_{<k}$ and $(B_A)_{<k}$. At any point $x \in D \setminus A$, simultaneous agreement with those two polynomials would force both $XB_A(x) = 0$ and $B_A(x) = 0$, impossible. Hence (f, g) is globally CA-far at agreement a, and the $n - a + 1$ slopes are CA-bad.

It remains to pass from finite to projective slopes. A projective bad-slope set cannot be all of $\mathbb{P}^1(F)$: otherwise, after choosing any projective point as infinity, the remaining q points would be finite bad slopes in an affine coordinate, contradicting $q > n-a+1$ since $D \subseteq F$ and $a > k$. Choose a nonbad projective point and send it to infinity by a change of basis of the two-dimensional received-line span. All bad projective points are then finite slopes in the new coordinate, so Theorem 7 bounds their number by $n-a+1$. The affine moving-root construction gives the matching projective lower bound. □

21.2 Finite-parameter curve MCA

Fix an integer $d \geq 1$. A degree-d received curve is a family

$$W_\gamma = f_0 + \gamma f_1 + \cdots + \gamma^d f_d, \qquad \gamma \in F.$$

Let $\mathrm{CurveLD}^{(d)}_{\mathrm{sw}}(C, a)$ be the maximum number of parameters $\gamma \in F$ for which W_γ has a C-explanation on a support of size at least a, while the tuple $(f_0, \ldots, f_d)$ is not simultaneously explained by C^{d+1} on that support. Define $\mathrm{CurveLD}^{(d)}_{\mathrm{ca}}(C, a)$ similarly, replacing the support-wise condition by the no-loss global CA condition that $(f_0, \ldots, f_d)$ is not C^{d+1}-close at agreement a.

Theorem 9 (degree-d curve tangent staircase)**.** *Let* $C = \mathrm{RS}[F, D, k]$*,* $|D| = n$*,* $|F| = q$*, and* $d \geq 1$*. If*

$$(d + 2)a - (d + 1)n \geq k,$$

then

$$\mathrm{CurveLD}^{(d)}_{\mathrm{sw}}(C, a) = \mathrm{CurveLD}^{(d)}_{\mathrm{ca}}(C, a) = \min\{q,\ d(n - a + 1)\}.$$

Proof. Put $m = n - a + 1$. For the lower bound, choose $A \subset D$ with $|A| = a - 1$ and let $\Omega = D \setminus A$, so $|\Omega| = m$. Choose $M = \min\{q, dm\}$ distinct parameters in F, partition them among the points $x \in \Omega$ into sets R_x of size at most d, and for each $x \in \Omega$ choose a nonzero polynomial

$$P_x(T) = \sum_{i=0}^{d} p_{i,x} T^i$$

whose root set contains R_x. Define $f_i = 0$ on A and $f_i(x) = p_{i,x}$ for $x \in \Omega$. If $\gamma \in R_x$, then W_γ vanishes on $A \cup \{x\}$, so it is explained by the zero codeword on a support of size a. On this same support the tuple $(f_0, \dots, f_d)$ is not simultaneously explained: every component vanishes on A, which has at least k points, so the only possible explaining codewords are all zero, but P_x is not the zero polynomial and hence some component is nonzero at x.

The same construction is CA-far. If the tuple were globally explained on a support T of size at least a, then

$$|T \cap A| \geq 2a - n - 1.$$

The curve-range hypothesis implies $2a - n - 1 \geq k$ when $a < n$, and the case $a = n$ is immediate from $n - 1 \geq k$. Thus all explaining codewords would be zero, which is impossible on every point of Ω by construction. Hence all M parameters are curve-CA-bad and support-wise curve-MCA-bad.

For the upper bound, it is enough to prove the support-wise statement, since curve-CA-bad parameters are support-wise bad. If there are at most d bad parameters, the bound is immediate. Otherwise pick $d + 1$ distinct bad parameters $\gamma_0, \dots, \gamma_d$, with explaining codewords on supports $S_0, \dots, S_d$. Their intersection has size at least

$$b_0 \geq (d + 1)a - dn.$$

On this intersection, interpolation in the parameter γ expresses each component f_i as an F-linear combination of the $d + 1$ explaining codewords, hence as a codeword $c_i \in C$. Let S_* be a maximal support on which all $f_i = c_i$, and put $b = |S_*|$. Then $b \geq b_0$, and the hypothesis gives

$$a + b - n \geq a + b_0 - n = (d + 2)a - (d + 1)n \geq k.$$

Now consider any bad parameter γ, with explaining codeword p_γ on a support S of size at least a. On $S \cap S_*$, whose size is at least $a + b - n \geq k$, the codeword p_γ agrees with

$$c_\gamma = \sum_{i=0}^{d} \gamma^i c_i \in C.$$

Therefore the residual word $W_\gamma - c_\gamma$ must vanish on at least $h = \max\{1, a - b\}$ points of $D \setminus S_*$. By maximality of S_*, for each residual coordinate $x \in D \setminus S_*$ the polynomial

$$R_x(T) = \sum_{i=0}^{d} T^i (f_i(x) - c_i(x))$$

is not identically zero, so it has at most d roots in F. Counting incidences (γ, x) with $R_x(\gamma) = 0$ gives

$$\#\{\text{bad parameters}\} \cdot h \leq d(n - b).$$

If $b \geq a$, then $h = 1$ and the right side is at most $d(n - a) \leq d(n - a + 1)$. If $b < a$, then

$$\#\{\text{bad parameters}\} \leq d\,\frac{n - b}{a - b} = d\left(1 + \frac{n - a}{a - b}\right) \leq d(n - a + 1),$$

because $a - b \geq 1$. The trivial field-size cap gives the additional upper bound q. This proves the exact formula. $\square$

21.3 Interleaved-list uniqueness

For $\mu \geq 1$, let $\Lambda_\mu(C, a)$ denote the maximum, over all μ-row received words $U = (U_1, \ldots, U_\mu)$, of the number of tuples $(c_1, \ldots, c_\mu) \in C^\mu$ that agree with U on at least a common positions.

Theorem 10 (high-agreement interleaved-list uniqueness)**.** *Let $C \subseteq F^D$ be MDS of length n and dimension k. If*

$$2a - n \geq k,$$

then for every $\mu \geq 1$,

$$\Lambda_\mu(C, a) = 1.$$

Proof. The lower bound is obtained by taking U itself to be a code tuple. For the upper bound, suppose two tuples $c = (c_1, \ldots, c_\mu)$ and $c' = (c'_1, \ldots, c'_\mu)$ both agree with U on supports of size at least a. The supports intersect in at least $2a - n \geq k$ positions. On this intersection, each row c_i agrees with c'_i. Since C is MDS, two codewords that agree on k positions are equal. Thus $c_i = c'_i$ for every row i, so the tuples are identical. □

21.4 The $\mathbb{F}_{17^{32}}$, $n = 512$, $k = 256$ protocol-facing ledger

Corollary 5 (high-agreement adjacent ledgers for the 17^{32} row)**.** *Let $H \subset \mathbb{F}^*_{17^{32}}$ have order 512, and let*

$$C = \mathrm{RS}[\mathbb{F}_{17^{32}}, H, 256].$$

Put $q = 17^{32}$ and

$$B_* = \left\lfloor q/2^{128} \right\rfloor = 6.$$

Then the following statements hold.

1. *For every $a \geq 427$,*

$$\mathrm{LD}_{\mathrm{ca}}(C, a) = \mathrm{LD}_{\mathrm{sw,proj}}(C, a) = \mathrm{LD}_{\mathrm{sw}}(C, a) = 513 - a.$$

   Thus the affine/projective line numerator is 6 at $a = 507$ and 7 at $a = 506$.

2. *For every $\mu \geq 1$ and every $a \geq 384$,*

$$\Lambda_\mu(C, a) = 1.$$

3. *For degree-d finite-parameter curves, whenever*

$$a \geq \left\lceil \frac{(d+1)512 + 256}{d+2} \right\rceil,$$

   we have

$$\mathrm{CurveLD}^{(d)}_{\mathrm{sw}}(C, a) = \mathrm{CurveLD}^{(d)}_{\mathrm{ca}}(C, a) = \min\{17^{32},\, d(513 - a)\}.$$

Proof. The first item is Corollary 3 together with Theorem 8. The second item is Theorem 10, because $2a - 512 \geq 256$ is equivalent to $a \geq 384$. The third item is Theorem 9 after substituting $n = 512$ and $k = 256$. The integer $B_* = 6$ follows from

$$6 \cdot 2^{128} < 17^{32} < 7 \cdot 2^{128}.$$

□

Corollary 6 (conditional high-agreement protocol ledger)**.** *Consider a protocol reduction for the row of Corollary 5 whose coding-theoretic error at agreement a is exactly*

$$\frac{N_{\mathrm{geom}}(a)}{17^{32}} + \frac{\Lambda_\mu(C,a)}{17^{32}} + \varepsilon_{\mathrm{query}},$$

where $N_{\mathrm{geom}}(a)$ is one affine/projective line CA/MCA numerator or one degree-d finite-parameter curve numerator, and where no additional fold, hash, Fiat–Shamir, or extension-lift loss is hidden in the notation. In the high-agreement ranges above, this ledger becomes the following explicit integer test.

For affine/projective line CA/MCA,

$$N_{\mathrm{geom}}(a) + \Lambda_\mu(C,a) = (513-a)+1 = 514-a \qquad (a \geq 427).$$

With $\varepsilon_{\mathrm{query}} = 0$, this is at most the 2^{-128} budget iff $a \geq 508$. Thus the combined line-plus-list ledger is safe at agreement 508 and unsafe at agreement 507.

For degree-d finite-parameter curve MCA/CA,

$$N_{\mathrm{geom}}(a) + \Lambda_\mu(C,a) = d(513-a)+1$$

in the curve-staircase range. With $\varepsilon_{\mathrm{query}} = 0$, the safe grid points are exactly those satisfying

$$d(513-a)+1 \leq 6.$$

In particular:

d	*curve-plus-list safe grid*	*curve term alone safe grid*
1	$a \geq 508$ $(r \leq 4)$	$a \geq 507$ $(r \leq 5)$
2	$a \geq 511$ $(r \leq 1)$	$a \geq 510$ $(r \leq 2)$
3	$a = 512$ $(r = 0)$	$a \geq 511$ $(r \leq 1)$
4	$a = 512$ $(r = 0)$	$a = 512$ $(r = 0)$
5	$a = 512$ $(r = 0)$	$a = 512$ $(r = 0)$
6	*no safe grid point with the list term*	$a = 512$ $(r = 0)$
$d \geq 7$	*no safe grid point with the list term*	*no safe grid point for the curve term alone*

where $r = n - a$ is the integer Hamming radius.

Proof. By Corollary 5, the interleaved-list numerator is 1 throughout the displayed high-agreement range. The affine/projective line numerator is $513-a$, and the degree-d curve numerator is $d(513-a)$. Since

$$\left\lfloor 17^{32}/2^{128} \right\rfloor = 6,$$

the code-level numerator must be at most 6 when $\varepsilon_{\mathrm{query}} = 0$. The displayed inequalities are exactly the resulting integer conditions. ☐

Remark 7 (what this does and does not settle)**.** *Theorems 8, 9, and 10 are unconditional finite statements under the hypotheses printed above: the CA/projective/curve statements are Reed–Solomon evaluation-domain statements, while interleaved uniqueness is MDS. Corollary 6 is conditional on the protocol really consuming only the printed coding terms over the printed field. It does not cover reductions with additional curve families, extension-line lifts, folding losses, query errors, or cryptographic errors unless those terms are added to the ledger.*

22 General High-Agreement Ledgers Beyond the $\mathbb{F}_{17^{32}}$ Row

Status: New-local finite Reed–Solomon/MDS theorem package plus Conditional protocol ledger. This section generalizes the high-agreement tangent-star method away from the special field $\mathbb{F}_{17^{32}}$. The statements below are finite-grid coding statements. They do not use smoothness, except when one wants to combine them with the quotient-core and generated-field ledgers elsewhere in this repository. The line lower bound may be proved either by the RS locator construction of Theorem 6 or by the coordinate tangent-star construction below; the upper bounds use only the MDS zero-rigidity property and linearity of the code.

Throughout this section let $C = \mathrm{RS}[F, D, k]$ with $D \subset F$, $|D| = n$, and dimension k. The upper-bound arguments below use only the MDS zero-rigidity property of Reed–Solomon codes, so they apply verbatim to any linear MDS code for which the relevant received words and parameter sets are defined. Put

$$R = n - k, \qquad r = n - a.$$

Thus r is the integer Hamming radius and $a = n - r$ is the agreement threshold. We only discuss the high-agreement range $a \geq k + 1$, equivalently $0 \leq r \leq R - 1$.

22.1 A coordinate tangent star

The next construction is useful because it removes all dependence on the particular field $\mathbb{F}_{17^{32}}$, on multiplicative smoothness, and on the locator-polynomial model. It uses only coordinate values, linearity, and the MDS zero-rigidity of Reed–Solomon codes.

Lemma 4 (coordinate tangent-star lower bound)**.** *Let $C = \mathrm{RS}[F, D, k]$ with $D \subset F$. For every a with $k + 1 \leq a \leq n$,*

$$\mathrm{LD}_{\mathrm{sw}}(C, a) \geq n - a + 1.$$

If in addition $2a - n - 1 \geq k$, then the same construction gives

$$\mathrm{LD}_{\mathrm{ca}}(C, a) \geq n - a + 1.$$

Proof. Choose $A \subset D$ with $|A| = a - 1$, and write $T = D \setminus A$, so $|T| = n - a + 1$. Choose an injection $\theta : T \hookrightarrow F$, which exists because $T \subseteq D \subset F$. Define two received words $f, g : D \to F$ by

$$f(x) = g(x) = 0 \quad (x \in A), \qquad f(t) = -\theta(t), \quad g(t) = 1 \quad (t \in T).$$

For a point $t \in T$, the finite slope $z = \theta(t)$ gives $f + zg = 0$ on $A \cup \{t\}$, a support of size a. Hence this slope is explained by the zero codeword.

The same support is support-wise noncontained. Indeed, if both f and g were separately explained on $A \cup \{t\}$, then the explaining codeword for g would vanish on the $a - 1 \geq k$ points of A, hence would be the zero codeword by the MDS property. But $g(t) = 1$, a contradiction.

The pair (f, g) is also globally far from C^2 at agreement a. The same argument applied to the word g shows that no codeword of C agrees with g on any support of size a: such a support contains at least $a - |T| = 2a - n - 1$ points of A, and in the high-agreement applications below we will have $2a - n - 1 \geq k$. Alternatively, for the lower bound in the line case one may use the RS locator construction of Theorem 6, which gives the no-loss CA witness directly for $\mathrm{RS}[F, D, k]$. The displayed MCA lower bound holds unconditionally for $a \geq k + 1$, and the CA lower bound holds in the range $2a - n - 1 \geq k$, in particular throughout the exact staircase range $3a - 2n \geq k$. ☐

Remark 8 (scope)**.** *The support-wise MCA construction is coordinate-level. It uses MDS rigidity and the inclusion $D \subset F$. The no-loss CA part needs global farness; in the exact range below, the needed support-count condition follows from the staircase hypothesis.*

22.2 Uniform line, CA, and projective-slope threshold

Theorem 11 (uniform high-agreement line staircase)**.** *Let $C = \mathrm{RS}[F, D, k]$ with $|D| = n$, and let $a = n - r$. If*

$$3a - 2n \geq k \qquad \text{equivalently} \qquad r \leq \frac{n-k}{3},$$

then

$$\mathrm{LD}_{\mathrm{sw}}(C, a) = \mathrm{LD}_{\mathrm{ca}}(C, a) = \mathrm{LD}_{\mathrm{sw,proj}}(C, a) = r + 1 = n - a + 1.$$

Here $\mathrm{LD}_{\mathrm{sw,proj}}$ allows slopes in $\mathbb{P}^1(F)$.

Proof. The finite-slope support-wise upper bound is Theorem 7, whose proof uses only MDS zero-rigidity and the common residual budget. The lower bound is Lemma 4, or, for Reed–Solomon codes, Theorem 6. No-loss CA-bad slopes are a subset of support-wise MCA-bad slopes, while the same tangent-star witness gives $r + 1$ no-loss CA-bad slopes in the present range, so the CA numerator is also $r + 1$.

For projective slopes, suppose a projective bad set had size larger than $r + 1$. If there is a nonbad projective point, use it as the point at infinity; all projective bad points then become finite bad slopes, contradicting the finite-slope bound. If every point of $\mathbb{P}^1(F)$ is bad, choose any point as infinity. The remaining $|F|$ finite slopes are bad. Since $|F| \geq n > r + 1$, this again contradicts the finite-slope bound. Thus the projective numerator is also $r + 1$, and the finite lower-bound construction shows equality. □

22.3 Degree-d curve MCA/CA

For an integer $d \geq 1$, a finite-parameter power curve is a received family

$$W_\gamma = f_0 + \gamma f_1 + \cdots + \gamma^d f_d, \qquad \gamma \in F.$$

The support-wise curve-MCA predicate asks for parameters γ for which W_γ has an explanation on a support of size at least a, while the $(d+1)$-tuple $(f_0, \ldots, f_d)$ is not simultaneously explained on that same support. The no-loss curve-CA predicate replaces support-wise noncontainment by global farness of the $(d+1)$-tuple from C^{d+1}.

Theorem 12 (degree-d high-agreement curve staircase)**.** *Let $C = \mathrm{RS}[F, D, k]$ with $|D| = n$, and let $a = n - r$. If*

$$(d+2)a - (d+1)n \geq k \qquad \text{equivalently} \qquad r \leq \frac{n-k}{d+2},$$

then

$$\mathrm{CurveLD}^{(d)}_{\mathrm{sw}}(C, a) = \mathrm{CurveLD}^{(d)}_{\mathrm{ca}}(C, a) = \min\{|F|,\ d(r+1)\}.$$

Proof. First prove the lower bound. Choose $A \subset D$ with $|A| = a - 1$ and put $T = D \setminus A$, so $|T| = r+1$. Choose disjoint parameter sets $\Gamma_t \subseteq F$, $t \in T$, with $|\Gamma_t| \leq d$ and $\sum_t |\Gamma_t| = \min\{|F|, d(r+1)\}$. For each t, choose a nonzero polynomial

$$h_t(Y) = h_{t,0} + h_{t,1}Y + \cdots + h_{t,d}Y^d$$

of degree exactly d, with leading coefficient nonzero, that vanishes on all parameters in Γ_t. Define received words by

$$f_i(x) = 0 \quad (x \in A), \qquad f_i(t) = h_{t,i} \quad (t \in T), \qquad 0 \leq i \leq d.$$

For $\gamma \in \Gamma_t$, the curve word W_γ vanishes on $A \cup \{t\}$, hence is explained by the zero codeword on a support of size a. The tuple is not simultaneously explained on that support: on the $a-1 \geq k$ points of A, every explaining endpoint codeword would have to be zero, but at t the coefficient vector of h_t is not zero. This gives the support-wise curve-MCA lower bound.

The same construction is globally far in the exact range. The last endpoint f_d is zero on A and nonzero on every point of T. Any support of size a contains at least $a - |T| = 2a - n - 1$ points of A. The hypothesis $r \leq (n-k)/(d+2)$, with $d \geq 1$, implies $2a - n - 1 \geq k$. Therefore any codeword agreeing with f_d on such a support would have to be the zero codeword, but the support must contain at least one point of T, where $f_d \neq 0$. Thus the endpoint tuple is not C^{d+1}-close at agreement a, proving the no-loss curve-CA lower bound.

For the upper bound, it is enough to rule out more than $d(r+1)$ bad parameters, because the trivial cap is $|F|$. If a curve had more than $d(r+1)$ bad parameters, then in particular it would have at least $d+1$ bad parameters. Select $d+1$ distinct bad parameters $\gamma_0, \dots, \gamma_d$ and supports S_i of size at least a. Their intersection has size at least

$$(d+1)a - dn.$$

On this intersection the $d+1$ explained codewords interpolate, as functions of γ, to codewords $c_0, \dots, c_d \in C$ explaining the endpoints $f_0, \dots, f_d$. Let S_0 be a maximal common support for this endpoint tuple, put $b = |S_0|$, and set $m = n - b$. Then

$$m \leq (d+1)(n-a) = (d+1)r.$$

For any further bad parameter, its explaining codeword must equal $c_0 + \gamma c_1 + \cdots + \gamma^d c_d$, because

$$a + b - n \geq (d+2)a - (d+1)n \geq k.$$

Thus outside S_0 it needs at least $h = \max(1, a-b)$ zeros of the residual coordinate polynomials. By maximality of S_0, no residual coordinate polynomial is identically zero, and each has degree at most d, hence can be charged to at most d bad parameters. If $b \geq a$, then $m \leq r$ and the count is at most $dm \leq dr \leq d(r+1)$. If $b < a$, then $h = a - b = m - r$ and $r < m \leq (d+1)r$; the charging bound gives

$$\#\{\text{bad parameters}\} \leq \left\lfloor \frac{dm}{m-r} \right\rfloor \leq d(r+1).$$

The trivial parameter-space cap gives the additional upper bound $|F|$. Together with the lower bound this proves equality. $\square$

22.4 Interleaved-list uniqueness

Theorem 13 (uniform interleaved-list uniqueness). *Let $C \subseteq F^D$ be a linear MDS code of dimension k, and let $\mu \geq 1$. If*

$$2a - n \geq k \qquad \textit{equivalently} \qquad r \leq \frac{n-k}{2},$$

then every μ-row received word has at most one μ-row codeword tuple agreeing with it on at least a common coordinates. Since equality is attained by taking the received word itself to be a code tuple,

$$\Lambda_\mu(C, a) = 1.$$

Proof. If two μ-row codeword tuples agree with the same received word on supports S and S' of size at least a, then they agree with each other on $S \cap S'$, whose size is at least $2a - n \geq k$. Row by row, the MDS property forces the two codewords to be equal. Hence the tuples are equal. The lower bound $\Lambda_\mu(C, a) \geq 1$ is obtained by choosing the received word to be a codeword tuple. $\square$

22.5 General finite-grid threshold calculator

Let $\varepsilon_* = 2^{-128}$ and let Q be the denominator of the sampler or challenge field used by a particular ledger. Define the integer budget

$$B_Q = \lfloor \varepsilon_* Q \rfloor .$$

The high-agreement theorems above convert coding ledgers into integer arithmetic. For example, in the exact line range $r \le \lfloor R/3 \rfloor$, a single affine line, no-loss CA, or projective-line numerator is

$$N_{\mathrm{line}}(r) = r + 1.$$

Thus the line term alone is safe whenever $r + 1 \le B_Q$, and the first unsafe radius is $r = B_Q$, provided $B_Q \le \lfloor R/3 \rfloor$. If $B_Q > \lfloor R/3 \rfloor$, then the tangent-star method proves safety throughout the exact high-agreement range but does not locate the eventual threshold.

For one degree-d finite-parameter curve term in its exact range $r \le \lfloor R/(d+2) \rfloor$, the numerator is

$$N_{\mathrm{curve,d}}(r) = \min\{|F|, d(r+1)\}.$$

If all coding terms use the same denominator Q, a protocol ledger with curve terms of degrees $d_1, \ldots, d_s$ and ℓ independent interleaved-list terms has high-agreement numerator

$$N_{\mathrm{total}}(r) = \ell + \sum_{i=1}^{s} d_i(r+1),$$

valid for

$$r \le \min\left\{ \left\lfloor \frac{R}{2} \right\rfloor \text{ if } \ell > 0, \min_i \left\lfloor \frac{R}{d_i + 2} \right\rfloor \right\}.$$

The ledger is certified at target ε_* whenever

$$N_{\mathrm{total}}(r) \le B_Q.$$

With different denominators Q_i, one should not combine numerators; the correct test is the rational inequality

$$\sum_i \frac{N_i(r)}{Q_i} + \varepsilon_{\mathrm{query}} \le \varepsilon_*.$$

Corollary 7 (row-independent closed-form gates)**.** *Let $R = n - k$, $B_Q = \lfloor Q/2^{128} \rfloor$, and assume the relevant radius lies in the exact high-agreement range.*

1. *A single affine/projective line, no-loss CA, or support-wise MCA term is safe for every integer radius*

   $$r \le B_Q - 1.$$

   The first unsafe radius is $r = B_Q$, if $B_Q \le \lfloor R/3 \rfloor$.

2. *A single line/CA/MCA term plus one interleaved-list term over the same field is safe for*

   $$r \le B_Q - 2.$$

   The first unsafe radius is $r = B_Q - 1$, if $B_Q - 1 \le \lfloor R/3 \rfloor$.

3. A single degree-d curve term is safe for

$$r \le \left\lfloor \frac{B_Q}{d} \right\rfloor - 1.$$

The first unsafe radius is $r = \lfloor B_Q/d \rfloor$, *if this radius is at most* $\lfloor R/(d+2) \rfloor$.

4. A single degree-d curve term plus one interleaved-list term over the same field is safe for

$$r \le \left\lfloor \frac{B_Q - 1}{d} \right\rfloor - 1.$$

The first unsafe radius is $r = \lfloor (B_Q - 1)/d \rfloor$, *if this radius is in range.*

Negative right-hand sides mean that even radius zero is not certified for that ledger and denominator.

22.6 Prize-rate applicability

For the Proximity Prize rates $\rho \in \{1/2, 1/4, 1/8, 1/16\}$, with $k \le 2^{40}$, the exact line range reaches integer radii

$$r \le \frac{n-k}{3} = \frac{(1-\rho)n}{3} = \frac{1-\rho}{3\rho} k.$$

Thus, for a field with $|F| \approx 2^{\lambda}$, the high-agreement tangent gate can pin the line threshold only when roughly

$$2^{\lambda - 128} \lesssim \frac{n-k}{3}.$$

At the maximal allowed dimension $k = 2^{40}$, this gives the following approximate largest field-bit sizes for which the line threshold can still be pinned inside the exact tangent range:

ρ	$(n-k)/3$ at $k = 2^{40}$	λ up to about
$1/2$	$2^{40}/3$	166.4
$1/4$	2^{40}	168.0
$1/8$	$7 \cdot 2^{40}/3$	169.2
$1/16$	$5 \cdot 2^{40}$	170.3

For substantially larger challenge fields, such as 2^{192} or 2^{256}, the tangent numerator is below the 2^{-128} budget throughout the exact high-agreement range for all prize-size rows. The threshold, if any, must then be controlled by the lower-agreement quotient-core, generated-field entropy, curve/folding, or aperiodic local-limit ledgers; the tangent-star method alone no longer pins it.

23 Towards-Prize Finite-Threshold Theorems

Status: New-local finite theorem infrastructure plus subsumed strict264 proof record. The results in this section are deliberately finite and certificate-facing. They are meant to move the experimental ledger from lower-bound numerators toward the integer agreement staircase needed by the Proximity Prize. They do not prove the corrected Paper B local-limit conjecture. The later high-agreement tangent staircase supersedes the old strict264/265 search target as a threshold-pinning route for the $\mathbb{F}_{17^{32}}$, $n = 512$, $k = 256$ row; the strict264 material below remains useful as certificate language and as a record of quotient-core mechanisms.

For a received line $\ell_z = f + zg$, write

$$\operatorname{Bad}_{f,g}(C, a)$$

for the set of finite slopes $z \in F$ for which there is a support $S \subseteq D$, $|S| \geq a$, and a codeword $c \in C$ satisfying $c|_S = (f + zg)|_S$, but there is no pair of codewords $c_f, c_g \in C$ with $c_f|_S = f|_S$ and $c_g|_S = g|_S$. Put

$$\mathrm{LD}^{f,g}_{\mathrm{sw}}(C, a) = |\operatorname{Bad}_{f,g}(C, a)|, \qquad \mathrm{LD}_{\mathrm{sw}}(C, a) = \max_{f,g} \mathrm{LD}^{f,g}_{\mathrm{sw}}(C, a),$$

where the maximum is over the line family allowed by the sampler under consideration. The field paying the denominator is always q_{line}, not silently q_{gen} or q_{chal}.

Proposition 8 (certificate-to-$\mathrm{LD}_{\mathrm{sw}}$ gate)**.** *Let $C = \mathrm{RS}[F, D, k]$, let $a \geq k$, and fix a received line $f + zg$. Suppose there is a finite set $\mathcal{C}$ of tuples*

$$(z_i, S_i, P_i), \qquad z_i \in F, \quad S_i \subseteq D, \quad |S_i| \geq a, \quad P_i \in F[X]_{<k},$$

with the following properties:

1. *the slopes z_i are pairwise distinct;*
2. *$P_i(x) = f(x) + z_i g(x)$ for every $x \in S_i$;*
3. *there are no $P_{i,f}, P_{i,g} \in F[X]_{<k}$ satisfying $P_{i,f}|_{S_i} = f|_{S_i}$ and $P_{i,g}|_{S_i} = g|_{S_i}$.*

Then

$$\mathrm{LD}^{f,g}_{\mathrm{sw}}(C, a) \geq |\mathcal{C}|, \qquad \mathrm{LD}_{\mathrm{sw}}(C, a) \geq |\mathcal{C}|.$$

Proof. Each tuple certifies that the slope z_i is explained on a support of size at least a, and the third condition is exactly support-wise noncontainment on that support. Pairwise distinctness of the slopes prevents multiple tuples from paying for the same bad slope. Thus all z_i's lie in $\operatorname{Bad}_{f,g}(C, a)$, giving the first inequality. The second follows from the definition of $\mathrm{LD}_{\mathrm{sw}}(C, a)$ as a maximum over lines. ☐

Corollary 8 (strict264 certificate target)**.** *For the row*

$$C = \mathrm{RS}[\mathbb{F}_{17^{32}}, H, 256], \qquad |H| = 512,$$

a strict264 certificate consists of seven tuples in Proposition 8 with $a = 264$. Such a certificate proves

$$\mathrm{LD}_{\mathrm{sw}}(C, 264) \geq 7.$$

Proof. Apply Proposition 8 with $|\mathcal{C}| = 7$ and $a = 264$. ☐

Proposition 9 (fixed-locator unique-slope principle)**.** *Use the notation of Theorem 4. Fix $a \geq k$, put $t = a - k$, $j = n - a = r - t$, and fix a squarefree D-split locator L_T of degree j. Let*

$$A_T = H_{t,j}(u)\boldsymbol{\ell}_T, \qquad B_T = H_{t,j}(v)\boldsymbol{\ell}_T.$$

If $B_T \neq 0$, then this locator contributes at most one finite support-wise noncontained slope. More precisely, a slope exists exactly when A_T is a scalar multiple of B_T, in which case the unique slope is

$$z_T = -A_{T,m}/B_{T,m}$$

for any coordinate m with $B_{T,m} \neq 0$, and the same value must be obtained from every nonzero coordinate of B_T. Consequently

$$\mathrm{LD}^{f,g}_{\mathrm{sw}}(C, a) \leq \#\{T \subset D : |T| = n - a, L_T \text{ squarefree and } D\text{-split}, B_T \neq 0, A_T \in FB_T\}.$$

Proof. For the fixed locator, Theorem 4 gives the incidence equation

$$A_T + zB_T = 0$$

and the noncontainment condition $B_T \neq 0$. If $B_T \neq 0$, any solution z is forced by any coordinate where B_T is nonzero. Existence is exactly the consistency condition that all coordinates give the same scalar, equivalently $A_T \in FB_T$. Counting possible locators gives the final bound; different locators may still collide to the same slope, so the displayed quantity is an upper bound on distinct slopes. □

Remark 9 (why this helps the 265 upper bound). *For the current $n = 512, k = 256$ row, this proposition remains useful for duplicate-aware locator searches. The old proposed upper bound* $\mathrm{LD}_{\mathrm{sw}}(C, 265) \leq 6$ *is now known to be false by Theorem 6, which already gives* $\mathrm{LD}_{\mathrm{sw}}(C, 265) \geq 248$. *Future upper-bound searches should therefore start beyond the exact tangent gate, at agreement* 507 *and higher, or explicitly isolate non-tangent mechanisms.*

Theorem 14 (subfield confinement for base-valued lines). *Let $K \subseteq F$ be a subfield, let $D \subset K$, and let $C_F = \mathrm{RS}[F, D, k]$. Suppose $f, g : D \to K$, and let $S \subseteq D$ have $|S| \geq k$. If $z \in F \setminus K$ and there is a polynomial $P \in F[X]_{<k}$ with*

$$P(x) = f(x) + zg(x) \qquad (x \in S),$$

then there are polynomials $P_f, P_g \in K[X]_{<k}$ such that

$$P_f|_S = f|_S, \qquad P_g|_S = g|_S.$$

Consequently, for K-valued received lines and agreement $a \geq k$, every support-wise noncontained finite slope lies in K.

Proof. Choose a subset $S_0 \subseteq S$ of size k. Interpolate the values of f and g on S_0 to obtain unique polynomials $P_f, P_g \in K[X]_{<k}$. On S_0, the polynomial $P_f + zP_g$ agrees with P. Since both have degree less than k and agree on k points, they are equal as polynomials over F. For every $x \in S$, therefore,

$$0 = f(x) + zg(x) - P(x) = (f(x) - P_f(x)) + z(g(x) - P_g(x)).$$

Both coefficients in the final expression lie in K. Since $z \notin K$, the elements $1, z$ are linearly independent over K. Hence both coefficients vanish for every $x \in S$, proving the separate explanations of f and g. Thus any explained slope outside K is support-wise contained, so a noncontained slope must lie in K. □

Remark 10 (extension-field warning). *Theorem 14 is intentionally one-sided. It only treats K-valued received lines. It does not rule out genuinely F-valued residue-line witnesses, and therefore must not be used as an extension-field MCA theorem without an additional hypothesis forcing the line to be K-valued.*

Theorem 15 (seven-slope arithmetic gate over $\mathbb{F}_{17^{32}}$). *Let $q_{\text{line}} = 17^{32}$. If a row over $\mathbb{F}_{17^{32}}$ has* $\mathrm{LD}_{\mathrm{sw}}(C, 264) \geq 7$, *then at radius*

$$\delta = 1 - \frac{264}{512} = \frac{31}{64}$$

its support-wise MCA error is strictly larger than 2^{-128}. In particular, for the $n = 512, k = 256$ row, any strict264 seven-slope certificate proves

$$\varepsilon_{\mathrm{mca}}(C, 31/64) > 2^{-128}.$$

Proof. The MCA sampler over finite slopes has denominator $q_{\text{line}} = 17^{32}$. Seven bad slopes therefore give error at least $7/17^{32}$. The exact integer comparison is

$$7 \cdot 2^{128} - 17^{32} = 140649736861019983995159547014262328 31 > 0.$$

Thus $7/17^{32} > 2^{-128}$. Agreement 264 on a domain of size 512 corresponds to relative radius $1 - 264/512 = 31/64$, giving the claim. □

Theorem 16 (one-step staircase pinning criterion)**.** *Fix a finite-slope sampler of denominator q_{line} and a target $\varepsilon_* = 2^{-128}$. Let*

$$B(a) = \mathrm{LD}_{\mathrm{sw}}(C, a)$$

for the allowed line family, and let $\delta_a = 1 - a/n$. If, for some integer $a \geq k$,

$$B(a) > \varepsilon_* q_{\text{line}} \qquad \text{and} \qquad B(a+1) \leq \varepsilon_* q_{\text{line}},$$

then the closed-ball agreement staircase is unsafe at radius δ_a and safe on the next open interval below it, including the grid radius δ_{a+1}. Equivalently, the largest grid radius certified safe is δ_{a+1}, while the continuous supremum of certified safe radii is δ_a and is not attained under the closed endpoint convention.

Before the tangent staircase was isolated, the special case

$$B(264) \geq 7, \qquad B(265) \leq 6$$

would have pinned the finite row to an unsafe/safe transition between

$$\delta_{265} = 247/512 \qquad \text{and} \qquad \delta_{264} = 31/64.$$

For the $\mathbb{F}_{17^{32}}$, $n = 512$, $k = 256$ row, the premise $B(265) \leq 6$ is false by the moving-root tangent floor. The actual finite-slope staircase gate supplied by Corollary 3 is $B(506) = 7$, $B(507) = 6$.

Proof. The function $B(a)$ is nonincreasing in a, because raising the agreement threshold only removes valid supports. At radius δ_a, the closed-ball predicate allows agreement at least a, so the error is $B(a)/q_{\text{line}}$, which is larger than ε_* by assumption. At radius δ_{a+1}, the predicate requires agreement at least $a+1$, so the error is $B(a+1)/q_{\text{line}}$, which is at most ε_*.

More generally, any radius $\delta < \delta_a$ sufficiently close to δ_a has integer distance bound at most $n - a - 1$, hence agreement at least $a + 1$. Thus the closed endpoint at δ_a is the first unsafe grid step, while the safe side is immediately below that endpoint. For $q_{\text{line}} = 17^{32}$, the inequalities

$$6 \cdot 2^{128} < 17^{32} < 7 \cdot 2^{128}$$

convert the printed integer conditions into the two displayed error inequalities, and $1 - 265/512 = 247/512$, $1 - 264/512 = 31/64$. □

24 June 29 Finite-Slope and Interleaved-List PR Packets

This section records the mathematically useful parts of PRs #131–#135. The branches were not merged wholesale, because they were stale relative to the current Paper D v6 and public-site state. The integrated source files are the compact notes and verifiers listed in `experimental/notes/triage/pr-triage-2026-06-29.md`.

24.1 Boundary-off external-anchor normal form

PR #131, due to AllenGrahamHart, contributes a local M1 proof-program normal form. For a shadow $S \subset D$ of size $j-1$ and an external anchor $\beta \notin D$, the one-outside locator satisfies

$$\ell_{S,\beta} = \ell_S^+ - \beta\ell_S^0.$$

Therefore the Hankel landing matrix

$$M_S(\beta) = [H(u)\ell_{S,\beta} \quad H(v)\ell_{S,\beta}]$$

has columns affine-linear in β, and the rank-one condition is cut out by quadratic minors in β. Thus, for fixed S, either a nonzero quadratic minor gives at most two external anchors, or all minors vanish identically and the branch is a ruled rank-one pencil. The later same-slope/root-slice note now classifies this ruled $t = 2$ Hankel branch: it is inactive or fixed finite slope, and the fixed-slope case lifts to a boundary root-slice core $H_{3,j-1}(u + z_0v)\ell_S = 0$. Therefore after fixed-slope boundary slices are charged, each shadow has at most two active non-ruled external anchors. This is recorded as PROVED-LOCAL / RULED-BRANCH CLASSIFIED / PROOF-PROGRAM / AUDIT, not as a proof of M1.

24.2 $a = 507$ adjacent bridge route cut

PR #132, due to Scott Hughes, records a useful non-promotion result for the $\mathbb{F}_{17^{32}}, n = 512, k = 256$ row. The tangent-star theorem gives

$$\mathrm{LD}_{\mathrm{sw}}(C, a) = 513 - a \qquad (a \geq 427),$$

so in particular

$$\mathrm{LD}_{\mathrm{sw}}(C, 507) = 6.$$

Any bridge that converted an adjacent line-plus-interleaved-list "6+1 = 7" ledger into one additional same-denominator finite-slope support-wise event would imply $\mathrm{LD}_{\mathrm{sw}}(C, 507) \geq 7$, contradicting the exact tangent-star bound. Hence the adjacent arithmetic remains a separate BRIDGE-NEEDED ledger and is not a new MCA row.

24.3 Interleaved-list lower row at agreement 326

PR #133, due to Scott Hughes, belongs to the interleaved-list track, not the MCA slope track. It proves high-agreement uniqueness

$$\Lambda_\mu(C, a) = 1 \qquad (384 \leq a \leq 512)$$

and a support-occupancy route cut excluding seven witnesses for

$$373 \leq a \leq 383.$$

It also gives lower-bound constructions at agreements 291, 292, 320, and, most usefully for the board, a hybrid quotient-residual construction with

$$\Lambda_\mu(C, 326) \geq 7.$$

Since

$$6 \cdot 2^{128} < 17^{32} < 7 \cdot 2^{128},$$

this clears the separate interleaved-list numerical gate over denominator $|F| = 17^{32}$. The packet does not claim an MCA numerator, protocol soundness, ordinary list-decoding exactness, or an exact δ_C^*.

24.4 Random simple-pole finite-slope floors

PR #134, due to Vadim Avdeev, proves a general random simple-pole lower bound. For $C = \mathrm{RS}[\mathbb{F}, D, k]$, $|F| = q$, $|D| = n$, $\beta \in F \setminus D$, and $k < a \le n$, put

$$C_a = \binom{n}{a}, \qquad s = a - k, \qquad R_a = \sum_{v=0}^{a-k-1} \binom{a}{v}\binom{n-a}{v} q^{-v}.$$

Then there is a simple-pole line with at least

$$\left\lceil \frac{qC_a}{C_a + R_a q^s} \right\rceil$$

finite bad slopes at agreement a. In the row

$$F = \mathbb{F}_{17}[z]/(z^{32} - 3), \quad H = \langle z \rangle, \quad |H| = 512, \quad k = 256,$$

the verified floors are

a	finite bad-slope lower bound
257	17^{32}
258	17^{32}
259	$17^{32} - 68{,}904$
260	33,439,260,151,101,646,297,506,087,371,119,470.

The direction is a genuine simple-pole direction: it is not a degree-<256 codeword on any support of size > 256.

24.5 Coset-packet finite-slope floors for $a = 261, \ldots, 288$

PR #135, also due to Vadim Avdeev, continues the finite-slope floor beyond the range where the random simple-pole second-moment bound is useful. The construction uses full dyadic cosets $K_M \le H$, whose locators have the form

$$\prod_{x \in hK_M} (X - x) = X^M - h^M,$$

so unions of full K_M-cosets share low elementary coefficients. After fixing a small partial coset when needed, this produces common-prefix support families. A separating deep anchor then turns the corresponding numerator polynomials into distinct finite bad slopes.

For the same $\mathbb{F}_{17^{32}}, n = 512, k = 256$ row, the certified floors are

a	finite bad-slope lower bound
$261 \ldots 263$	$\binom{63}{32}$
264	$\binom{64}{33}$
$265 \ldots 271$	$\binom{31}{16}$
272	$\binom{32}{17}$
$273 \ldots 287$	$\binom{15}{8}$
288	$\binom{16}{9}$.

Every displayed row beats the moving-root tangent floor $513 - a$, but these are still lower-bound rows only. They do not prove a safe-side theorem, an ordinary list-decoding theorem, an interleaved-list statement, or protocol soundness.

25 Non-Claims

This note does not claim:

- a proof of the corrected MCA conjecture;
- a positive theorem above corrected reserve;
- a q_{gen} local-limit theorem;
- a list, CA, MCA, line-decoding, SNARK, or protocol consequence;
- a large $\Theta(p^2)$ counterpacket;
- a new prize-worthy numerator beyond the current Cycle116/119 52,747,567,092 finite/source-scoped record;
- priority for the BCHKS quotient-locator proof pattern.

The mathematical value is narrower. On the imported side, the file gives a provenance-safe dictionary and wrapper for using BCHKS without claiming it as new. On the local Paper B side, the possible large branch in the restricted quadratic-extension window must satisfy an explicit algebraic identity before it can even be surface-sized.

A BibTeX Entry to Copy into Main Papers

```
@misc{BCHKS25,
  author = {Eli Ben-Sasson and Dan Carmon and Ulrich Habock and
            Swastik Kopparty and Shubhangi Saraf},
  title  = {On Proximity Gaps for Reed--Solomon Codes},
  year   = {2025},
  note   = {Manuscript dated November 11, 2025},
  url    = {https://www.math.toronto.edu/swastik/rs-proximity-gaps-2025.pdf}
}
```

References

[1] The Proximity Prize. *The Proximity Prize: Reed–Solomon Challenge*. Preliminary version, accessed June 26, 2026. https://proximityprize.org/.

[2] Eli Ben-Sasson, Dan Carmon, Ulrich Haböck, Swastik Kopparty, and Shubhangi Saraf. *On Proximity Gaps for Reed–Solomon Codes*. Manuscript dated November 11, 2025. https://www.math.toronto.edu/swastik/rs-proximity-gaps-2025.pdf.